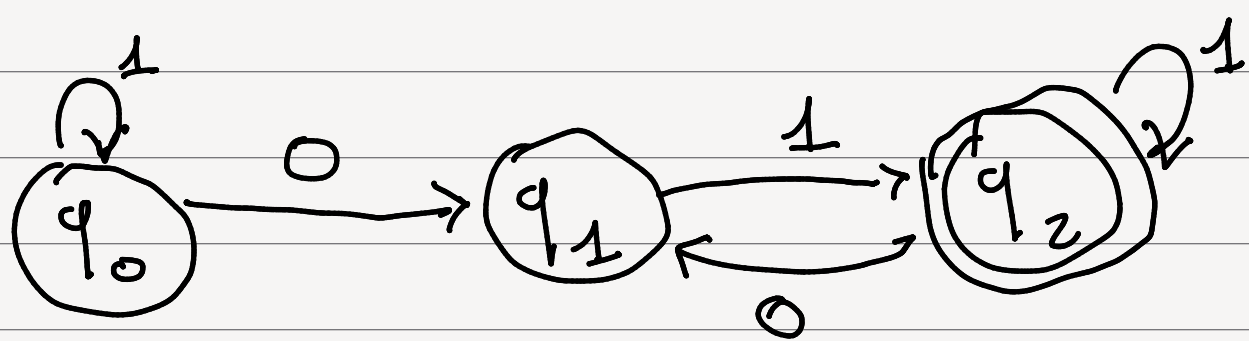
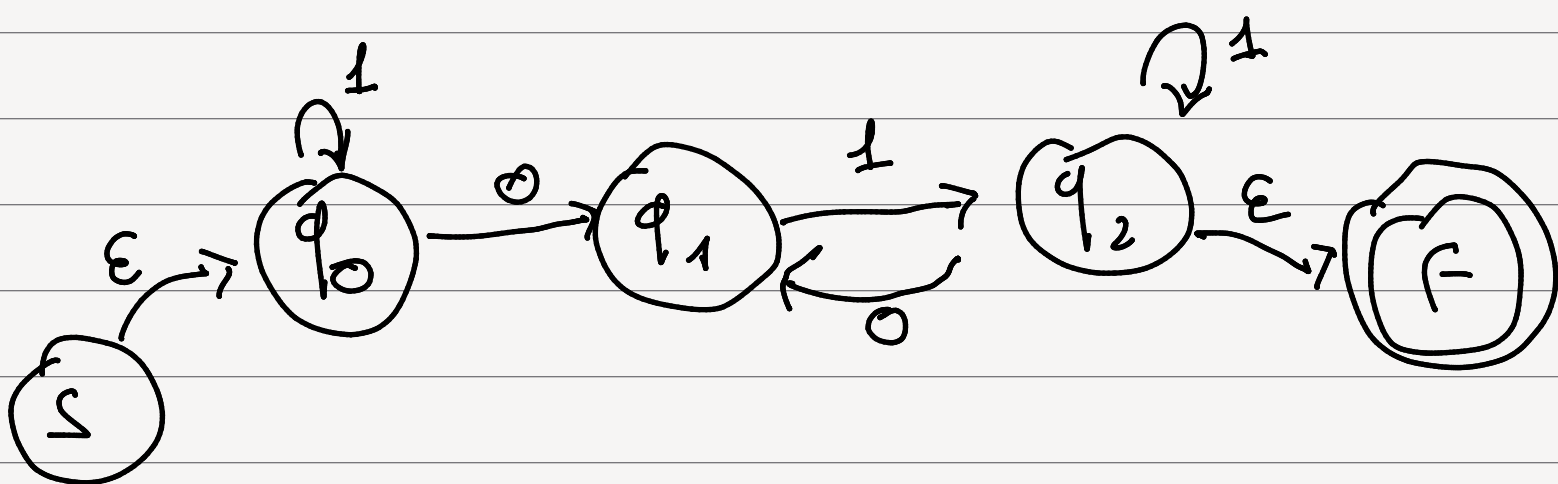


27. Stringhe binarie in cui ogni 0 ha almeno un 1 a destra



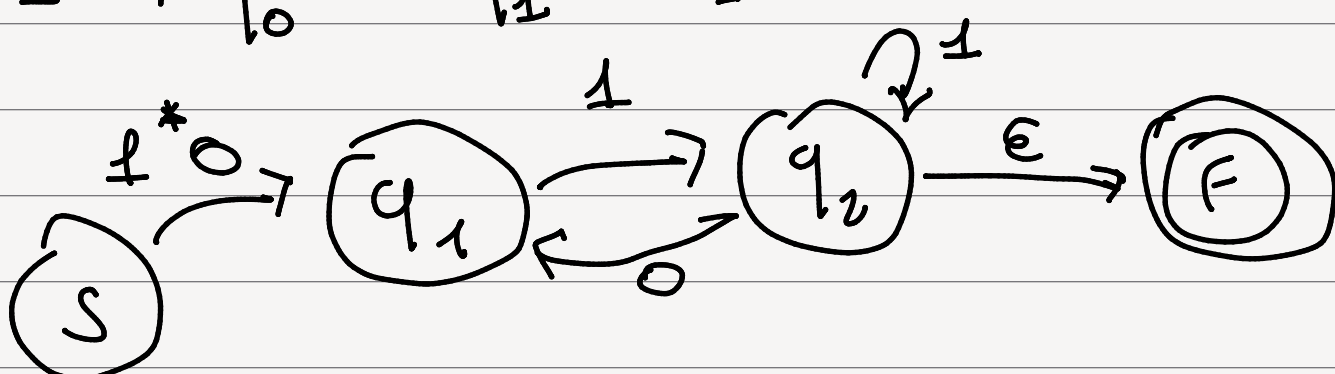
Trasformo in GNFA per la regex equivalente:



Elimino q_0 :

In	Out
S	q_1

$S \rightarrow q_0 \rightarrow q_1 = 1^* 0$



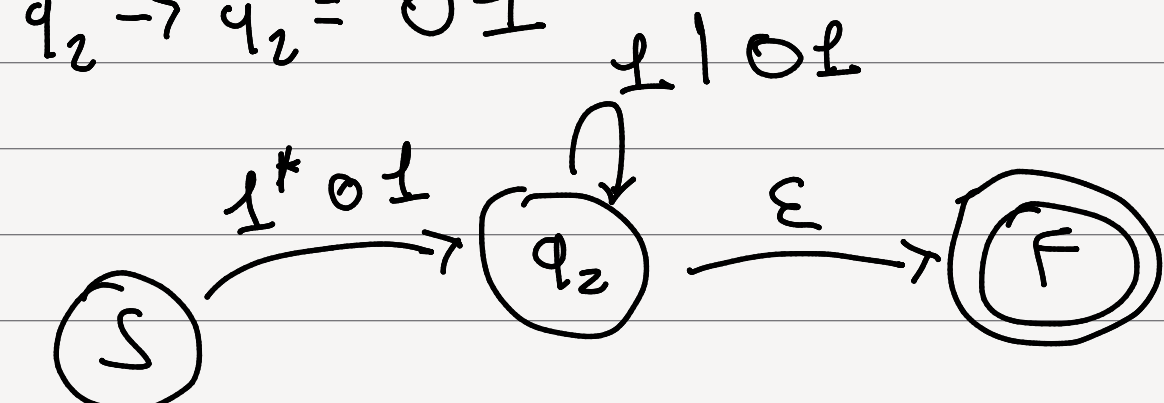
Elimino q_1 :

In	Out
S, q_2	q_2

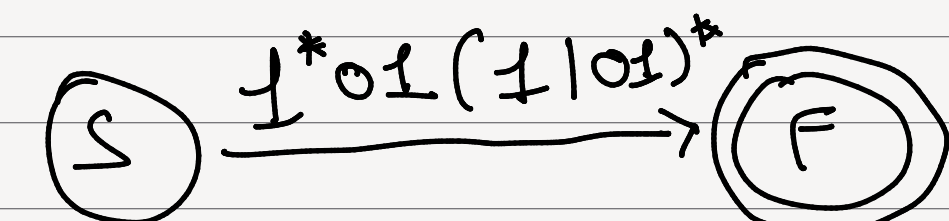
$S, q_2 \rightarrow q_2$

$S \rightarrow q_2 = 1^* 0 1$

$q_2 \rightarrow q_2 = 0 1$

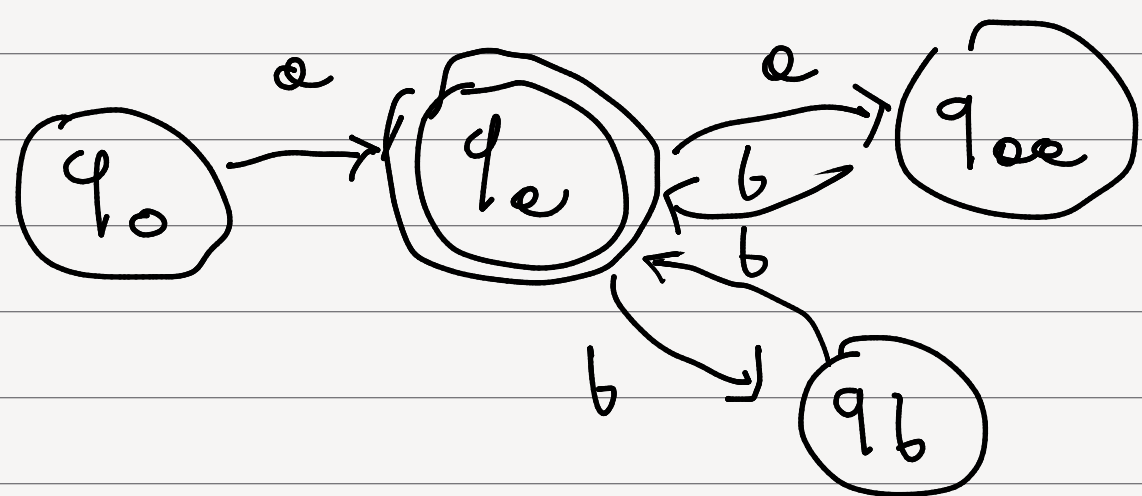


Elimino q_2 :

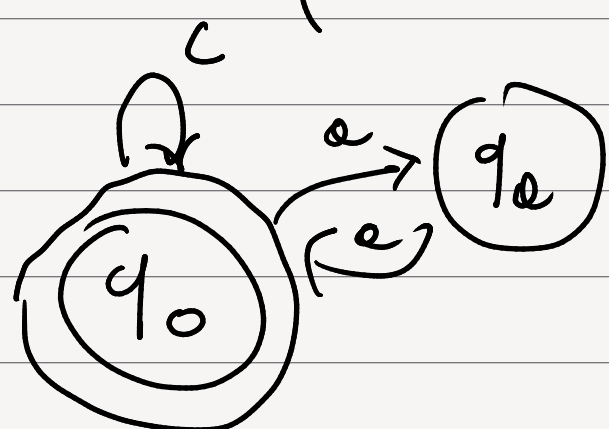


19. $a(ab|bb)^* \mid (aa|c)^*$

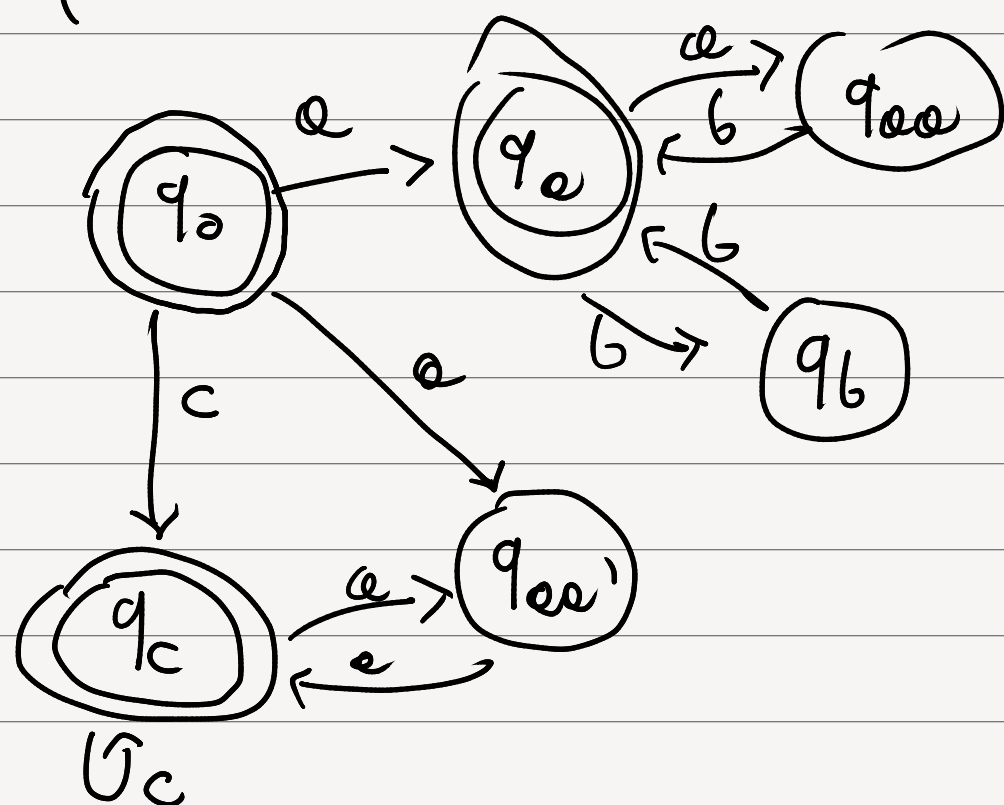
DFA per $a(ab|bb)^*$



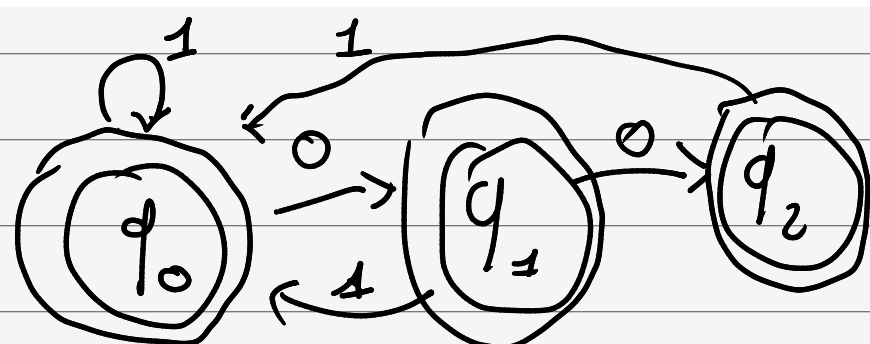
DFA per $(aa|c)^*$



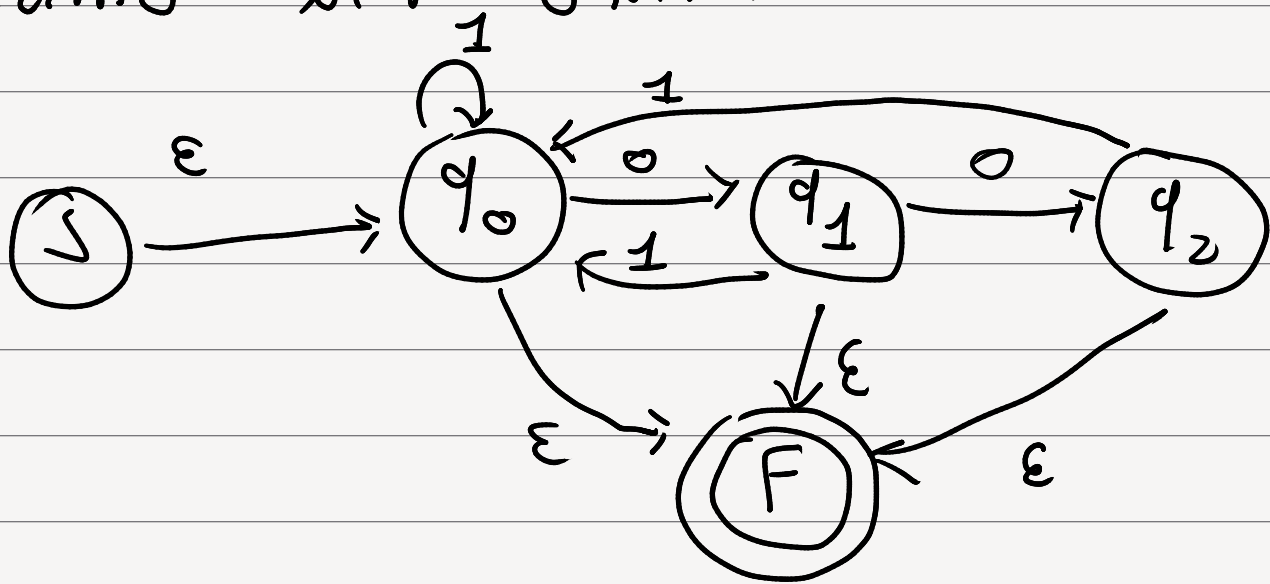
NFA per $a(ab|bb)^* \mid (aa|c)^*$



29. Stringhe binarie in cui non ci sono più di due 0 consecutivi



Transform to GNFA:

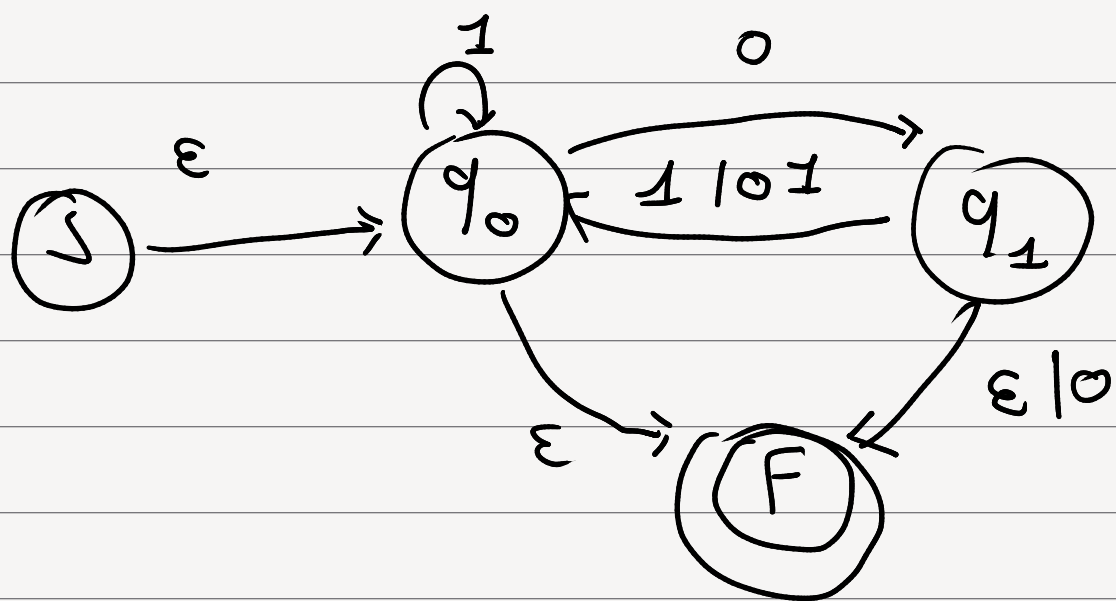


Path q_1 :

In Out
 q_1 q_0, F

$$q_1 \rightarrow q_0 = 01$$

$$q_1 \rightarrow F = 0$$



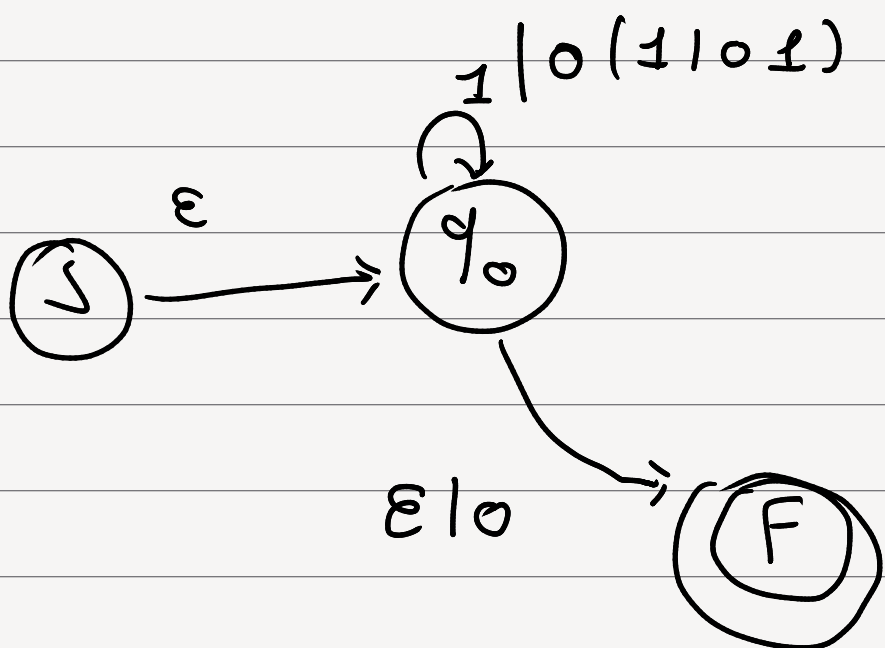
Path q_1

In Out

q_0 q_0, F

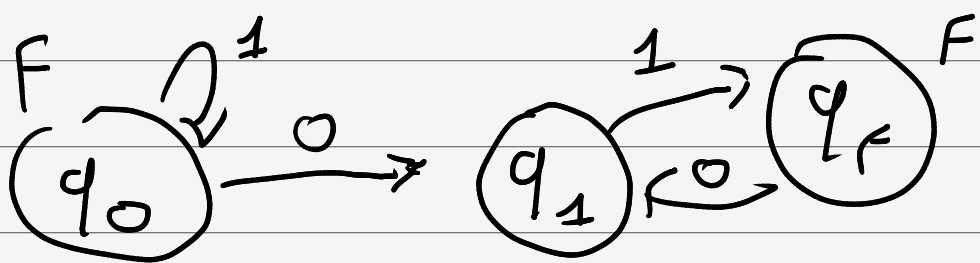
$$q_0 \rightarrow q_0 = 0(1|01)$$

$$q_0 \rightarrow F = \epsilon|0$$

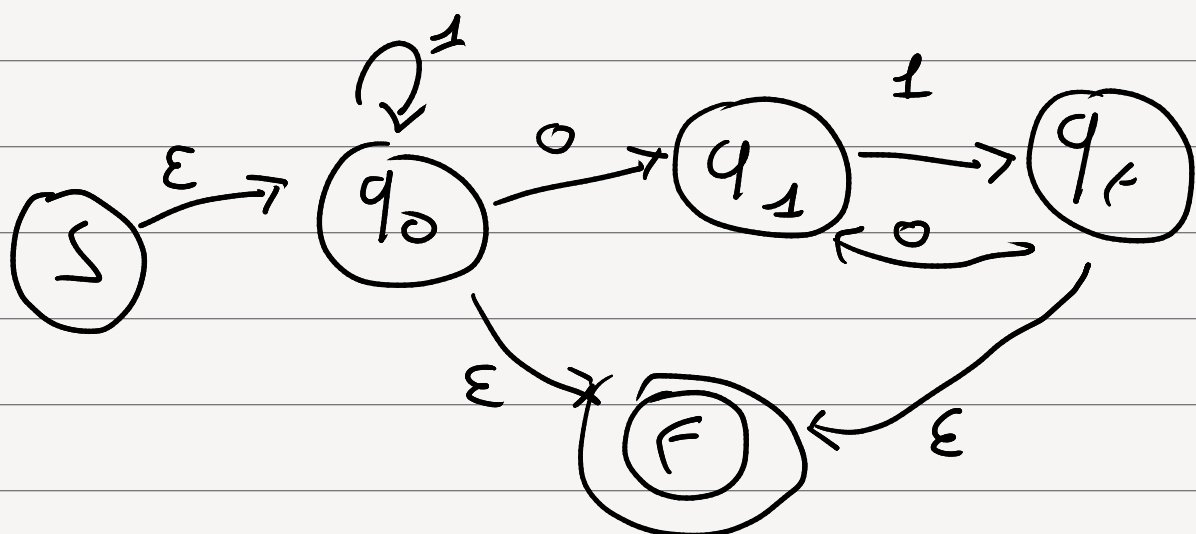


$$S \rightarrow F = (1|0(1|01))^* (\epsilon|0)$$

28. Stringhe binarie in cui ogni 0 ha esattamente un 1 a destra



GNFA:

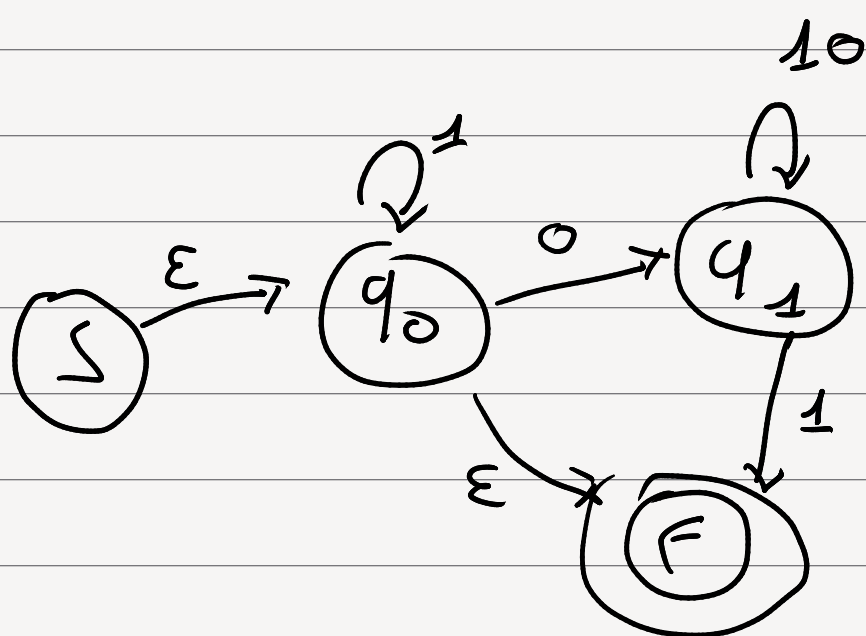


Tracce q_0 :

In Out
 q_1 q_1, F

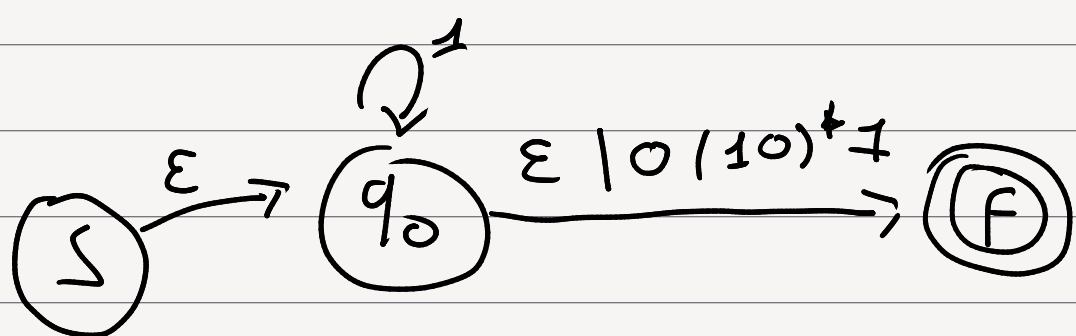
$q_1 \rightarrow q_1 = 10$

$q_1 \rightarrow F = 1$



Tracce q_1 :

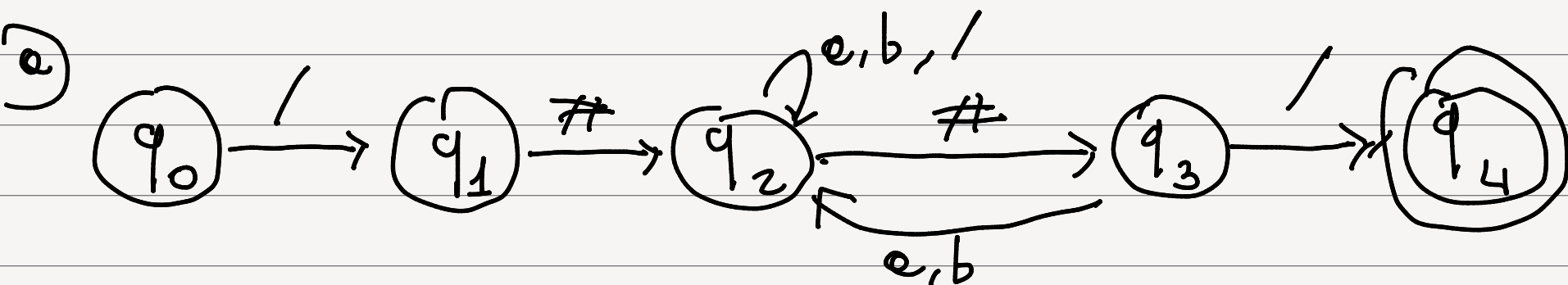
$q_0 \rightarrow F = \epsilon \mid 0(10)^*1$



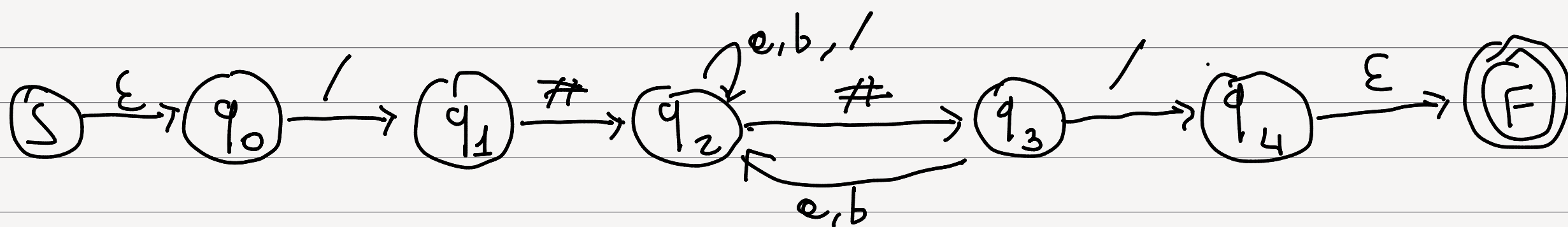
$S \rightarrow F = 1^* (\epsilon \mid 0(10)^*1)$

1.22 In alcuni linguaggi di programmazione, i commenti appaiono tra delimitatori come $/\#$ e $\#/$. Sia C il linguaggio di tutte le stringhe di commento correttamente delimitate. Un elemento di C deve iniziare con $/\#$ e terminare con $\#/$ ma non deve avere alcun $\#/$ intermedio. Per semplicità, assumere che l'alfabeto per C sia $\Sigma = \{a, b, /, \#\}$.

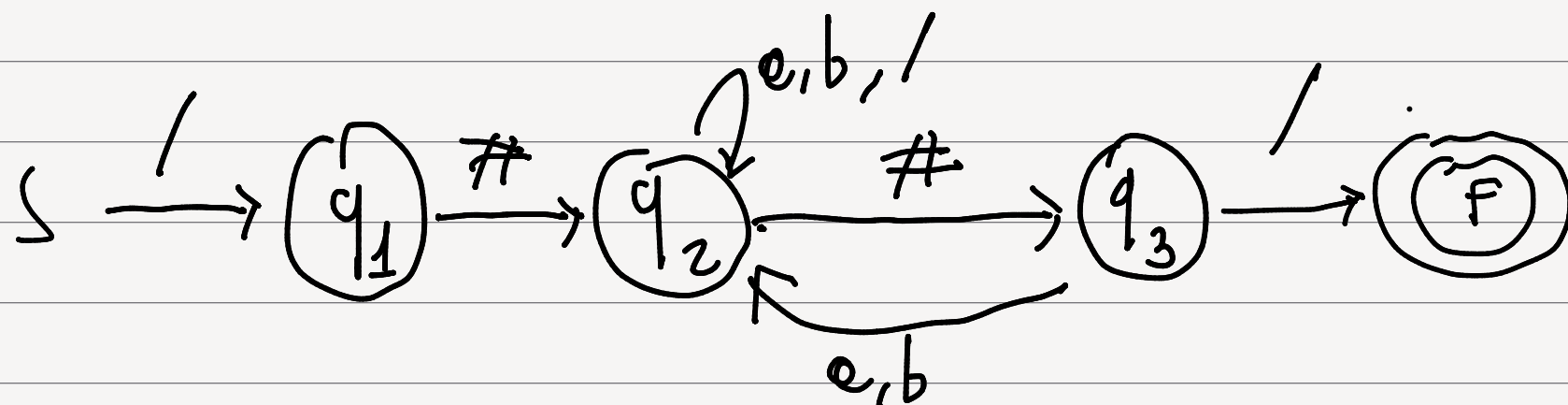
- Dare un DFA che riconosce C .
- Dare un'espressione regolare che genera C .



b) Costruiamo il GNFA che riconosce C .



che possiamo riscrivere come



Eliminiamo q_3 .

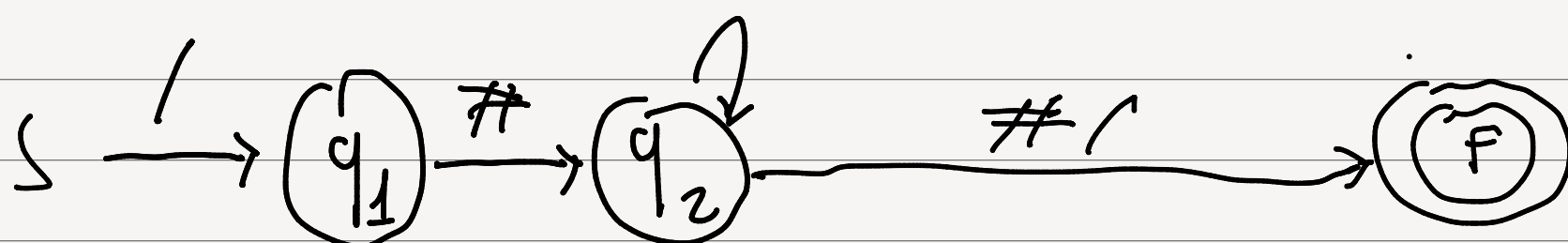
In Out

q_2 q_2, F

$q_2 \rightarrow q_2 = \#a \mid \#b$

$q_2 \rightarrow F = \# /$

$(a \mid b \mid /)^* (\#a \mid \#b)$

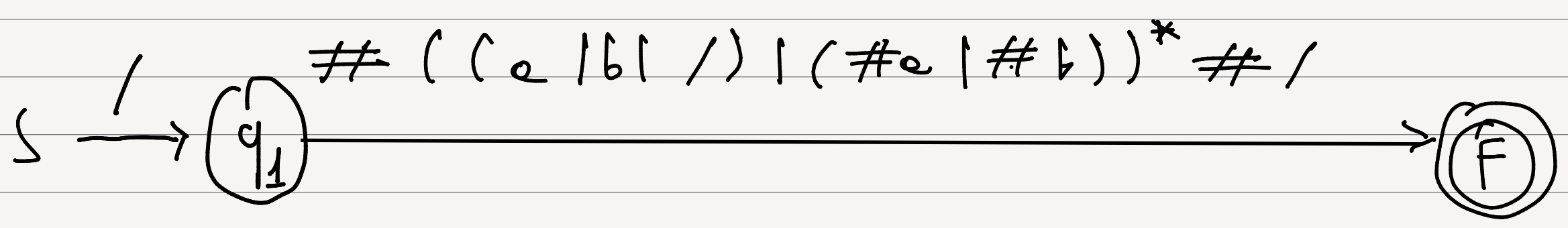


Eliminiamo q_2 :

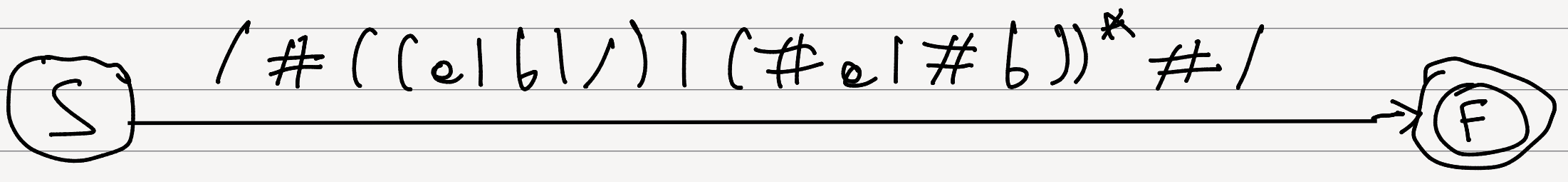
In Out

q_1 F

$$q_1 \Rightarrow F = \# ((a|b|/)|(\#a|\#b))^* \# /$$



che diventa:

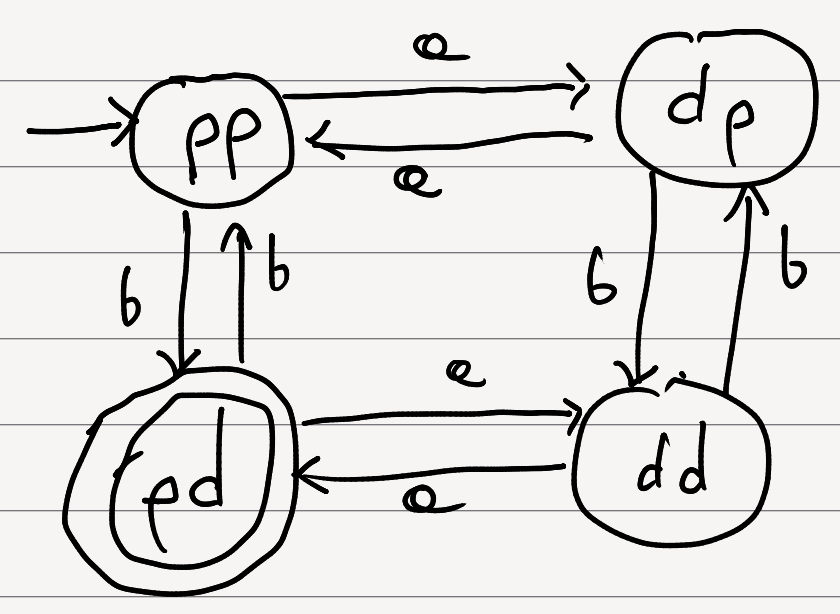


1.12 Sia $D = \{w \mid w \text{ contiene un numero pari di } a \text{ e un numero dispari di } b \text{ e non contiene la sottostringa } ab\}$. Fornire un DFA con cinque stati che riconosce D e un'espressione regolare che genera D . (Suggerimento: Descrivere D più semplicemente.)

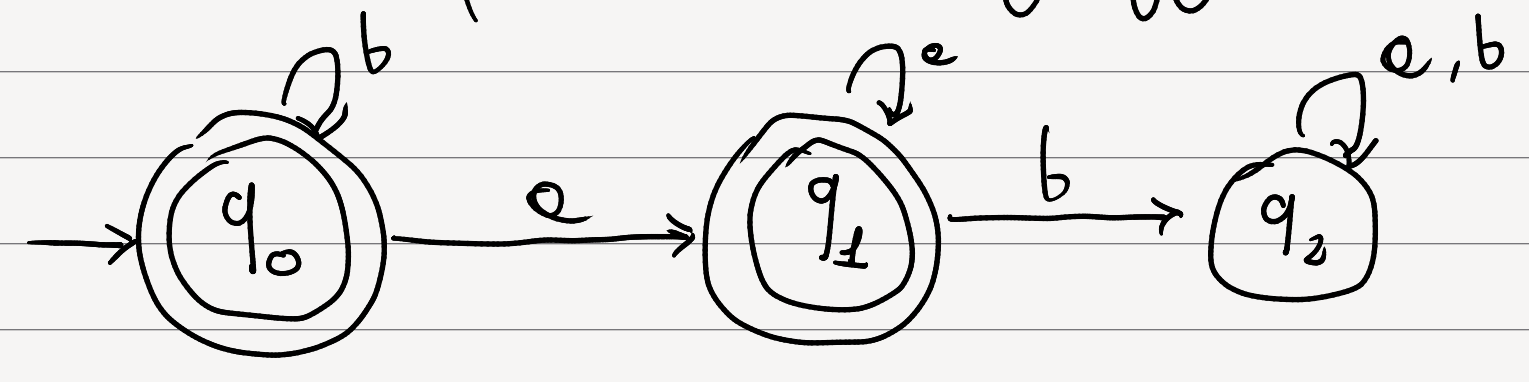
Il linguaggio D può essere visto come:

$$D = \{w \in \{a, b\}^+ \mid \#(a) = 2k, \#(b) = 2m + 1; k, m \geq 0\} \cap \{w \neq 2ab\}$$

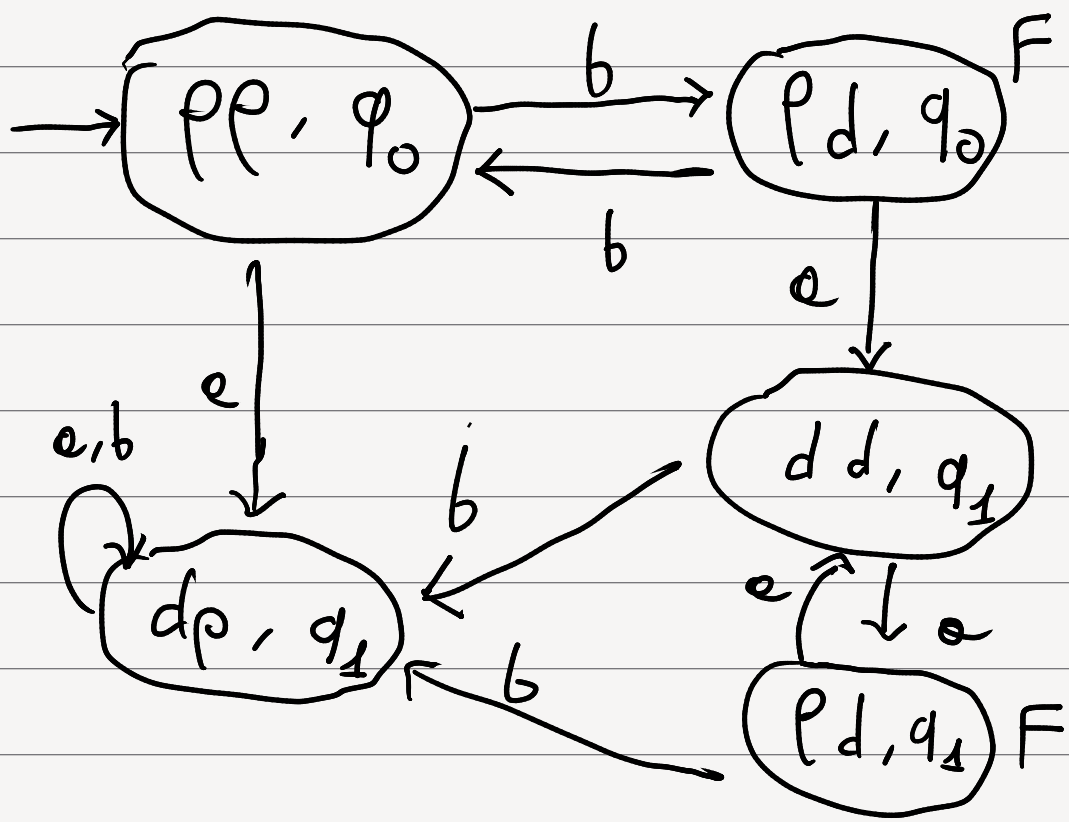
Il DFA per il linguaggio D_1 :



Il DFA per il linguaggio D_2 :

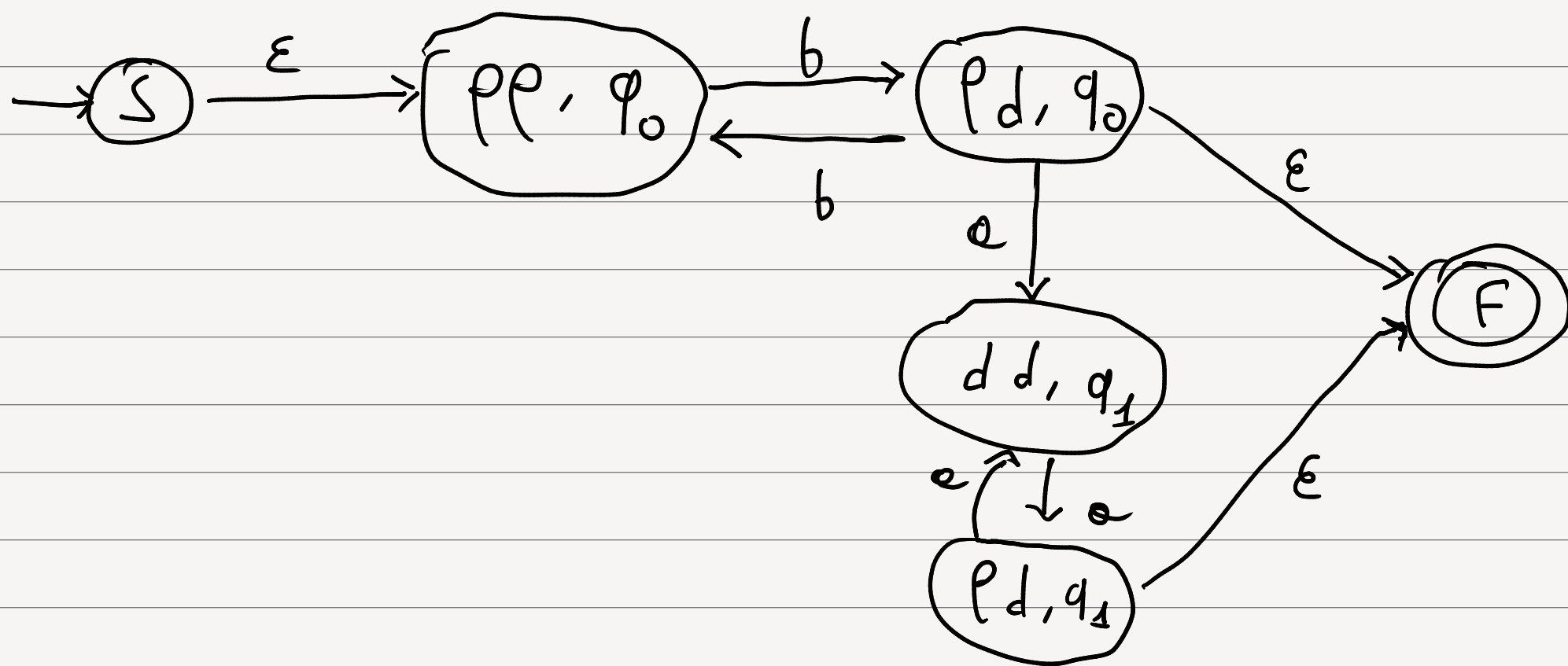


Costruiamo il DFA per D :



Adesso costruiamo il GNFA per D:

Eliminiamo il trap state dp, q_1

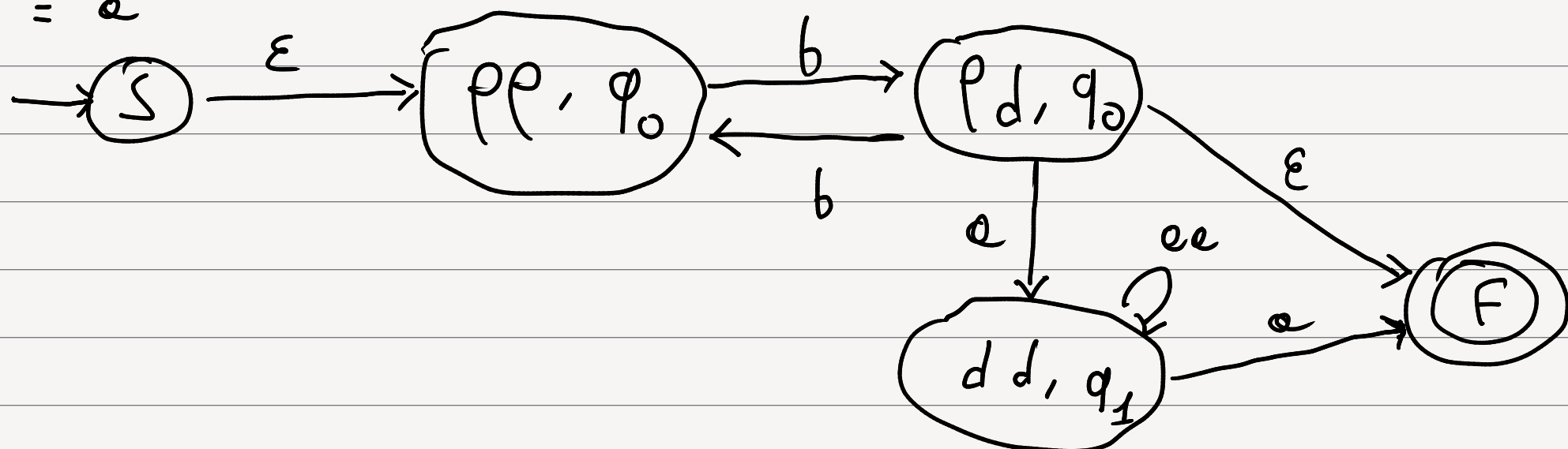


Eliminiamo pd, q_1

Im Out
 dd, q_1 dd, q_1
 f

$dd, q_1 \rightarrow dd, q_1 = ee$

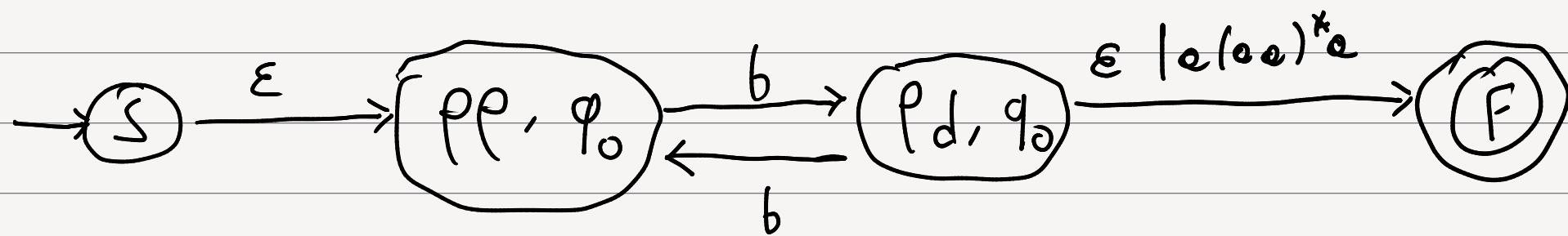
$dd, q_1 \rightarrow F = e$



Eliminiamo dd, q_1 :

Im Out
 pd, q_0 f

$$pd\ q_0 \rightarrow F = e(ee)^*e \mid \epsilon$$

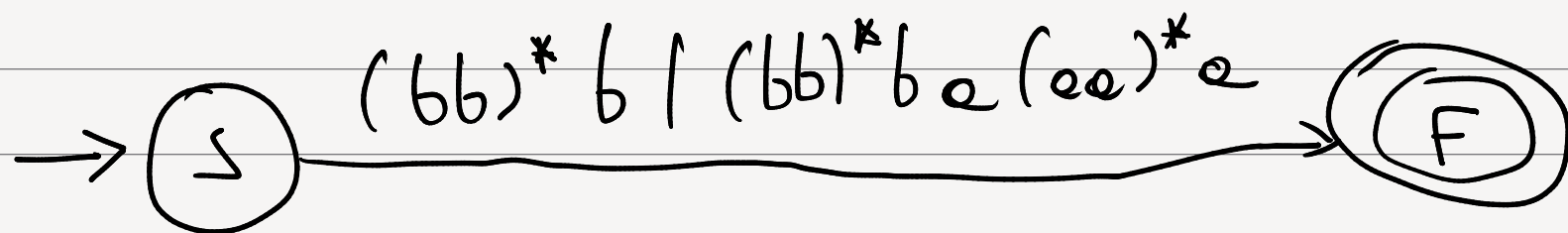
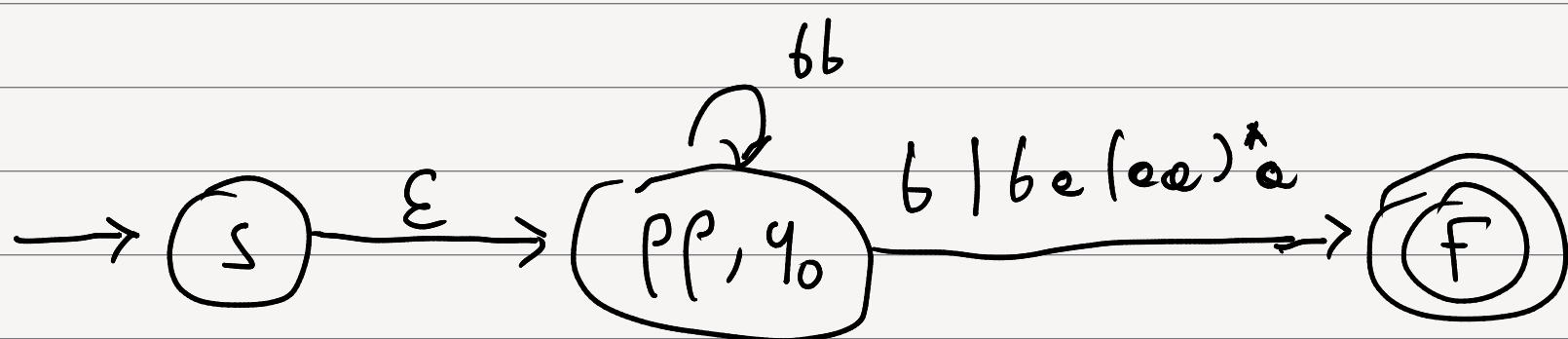


Eliminiamo pd, q_0 :

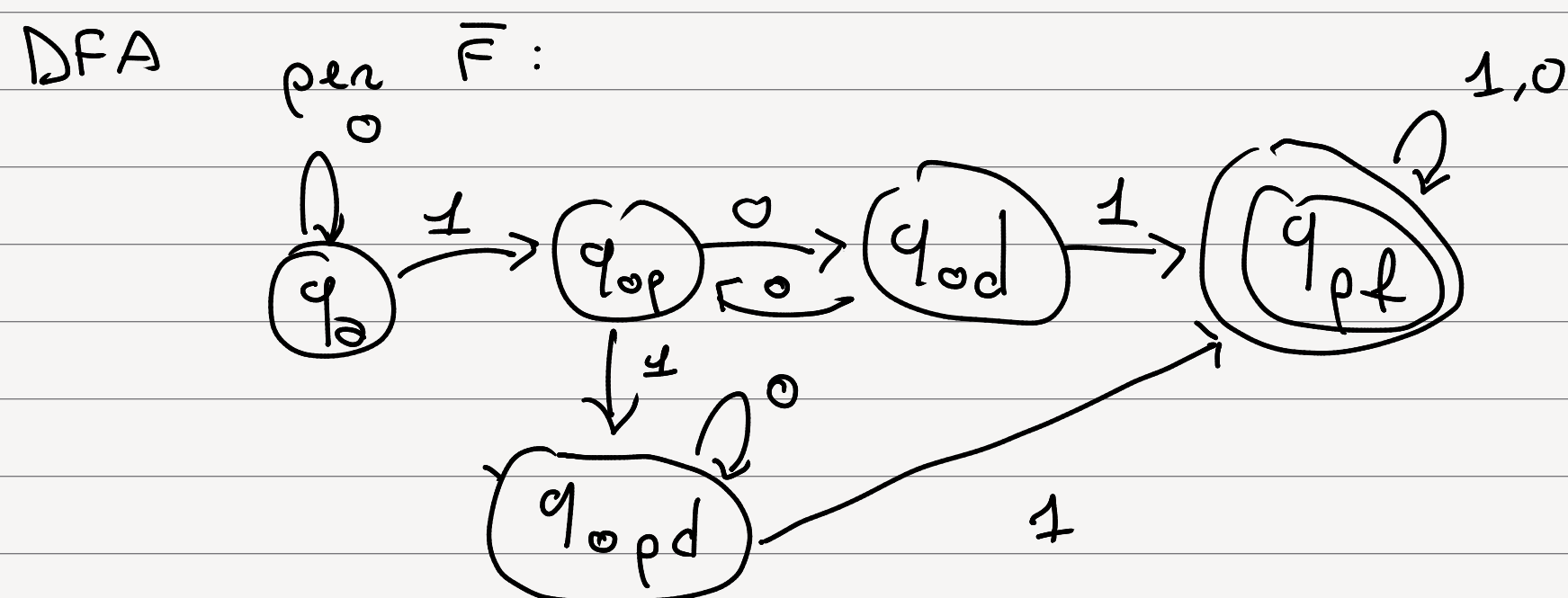
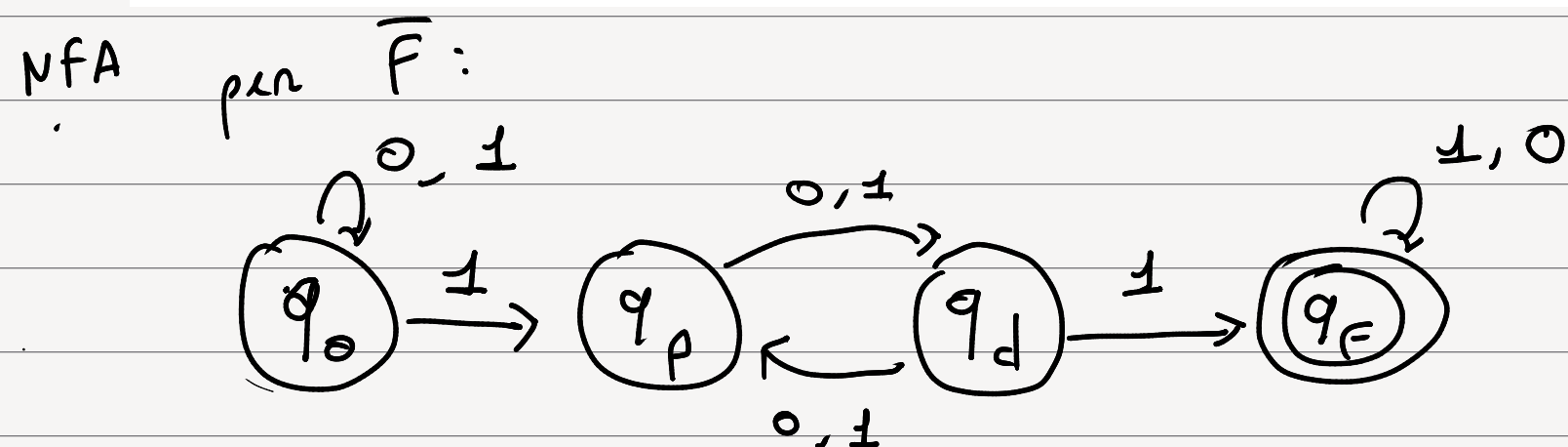
In Out
 ppq_0 ppq_0
 F

$$ppq_0 \rightarrow ppq_0 = bb$$

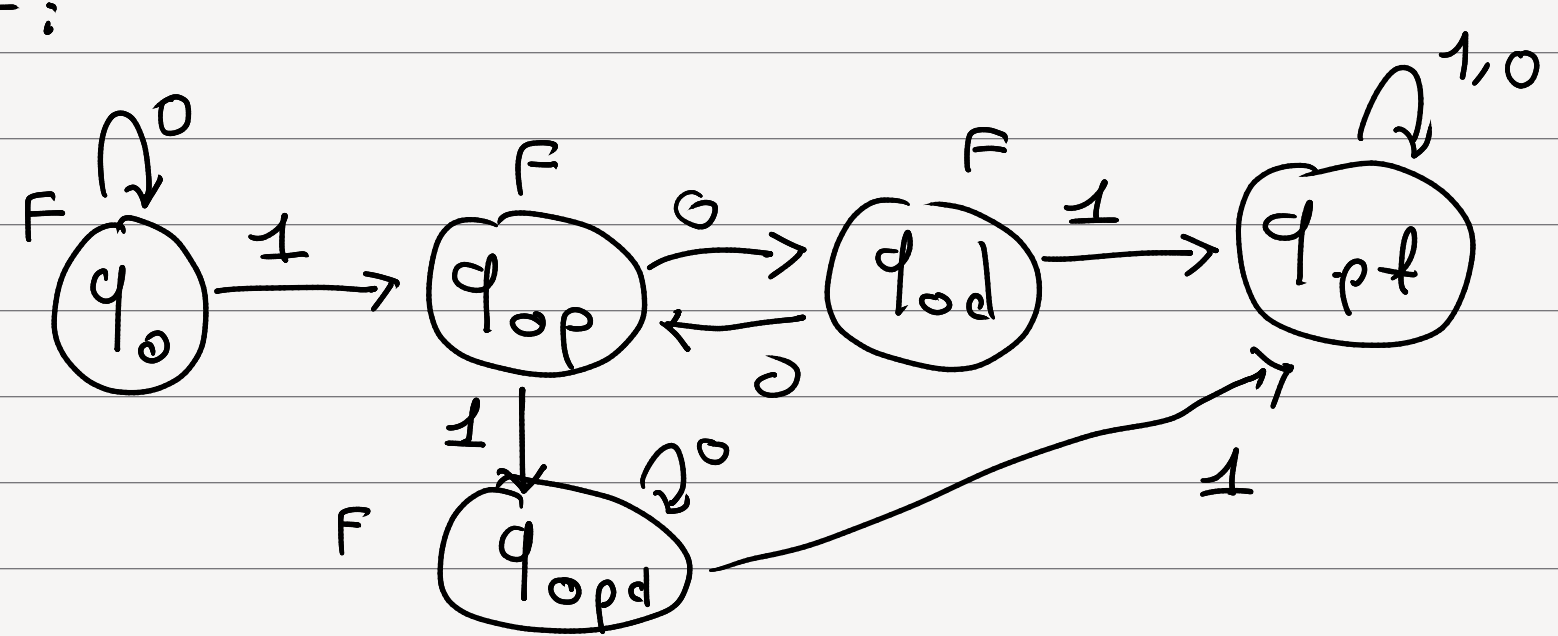
$$ppq_0 \rightarrow F = b \mid be(ee)^*e$$



1.13 Sia F il linguaggio di tutte le stringhe su $\{0,1\}$ che non contengono una coppia di simboli uguali a 1 separati da un numero dispari di simboli. Fornire il diagramma di stato di un DFA con cinque stati che riconosce F . (Potresti trovare utile trovare prima un NFA con 4 stati per il complemento di F .)



DFA per F :



Esercizio 2

Sia $M = (Q, \delta, q_0, F)$ un automa a stati finiti nondeterministico di alfabeto $X = \{0, 1\}$, ove:

$Q = \{q_0, q_1, q_2, q_3\}$,

$\delta(q_0, 0) = \{q_0\}$ $\delta(q_0, 1) = \{q_1, q_2\}$

$\delta(q_1, 0) = \{q_3\}$ $\delta(q_1, 1) = \{q_0, q_1\}$

$\delta(q_2, 0) = \{q_0, q_1\}$ $\delta(q_2, 1) = \{q_0, q_2\}$

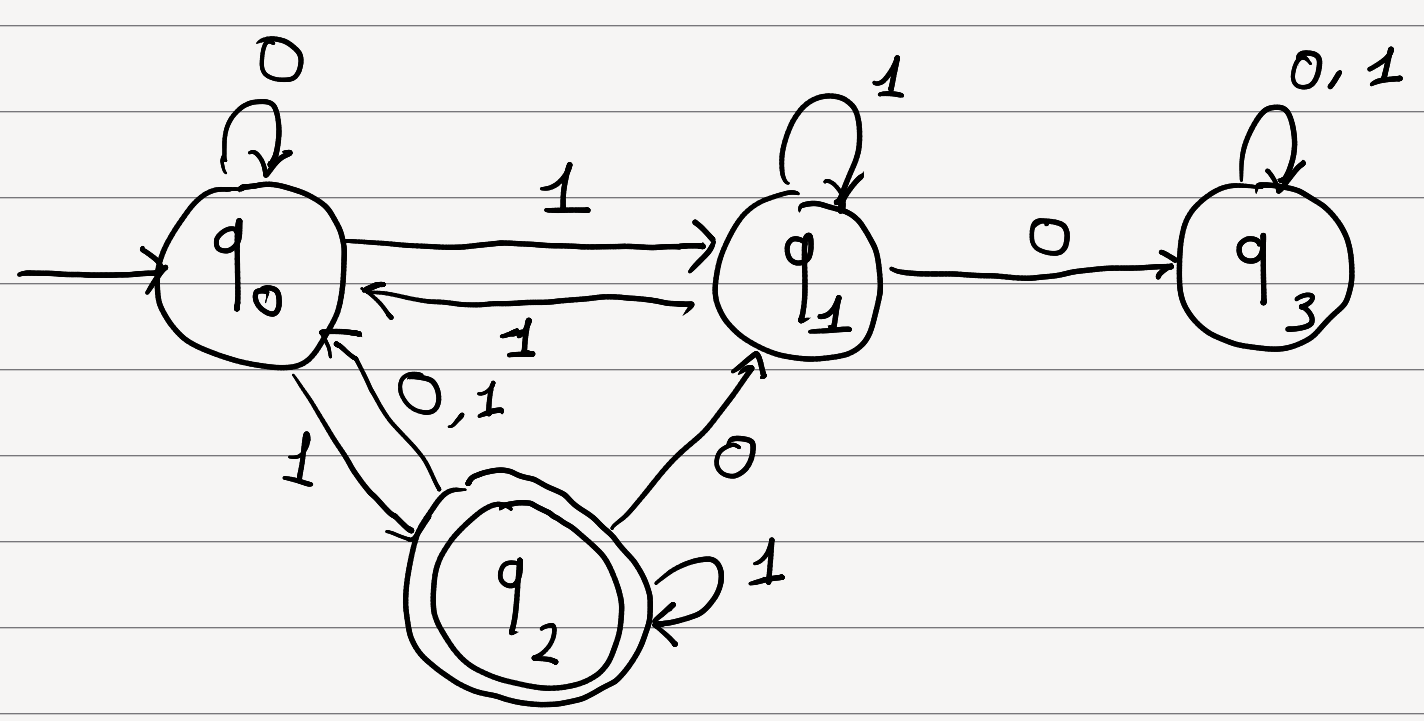
$\delta(q_3, 0) = \{q_3\}$ $\delta(q_3, 1) = \{q_3\}$

ed $F = \{q_2\}$

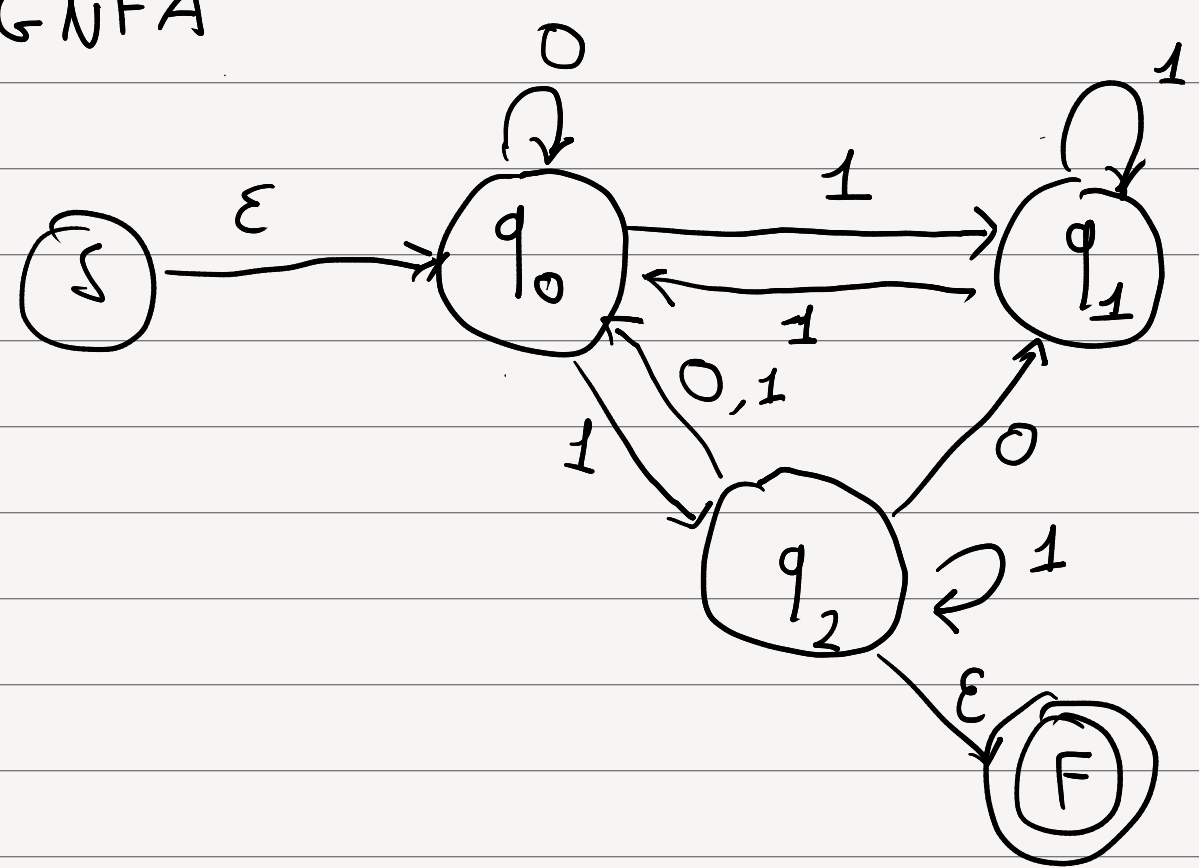
Descrivere il linguaggio accettato $T(M)$.

Costruire un automa a stati finiti deterministico che accetta $T(M)$.

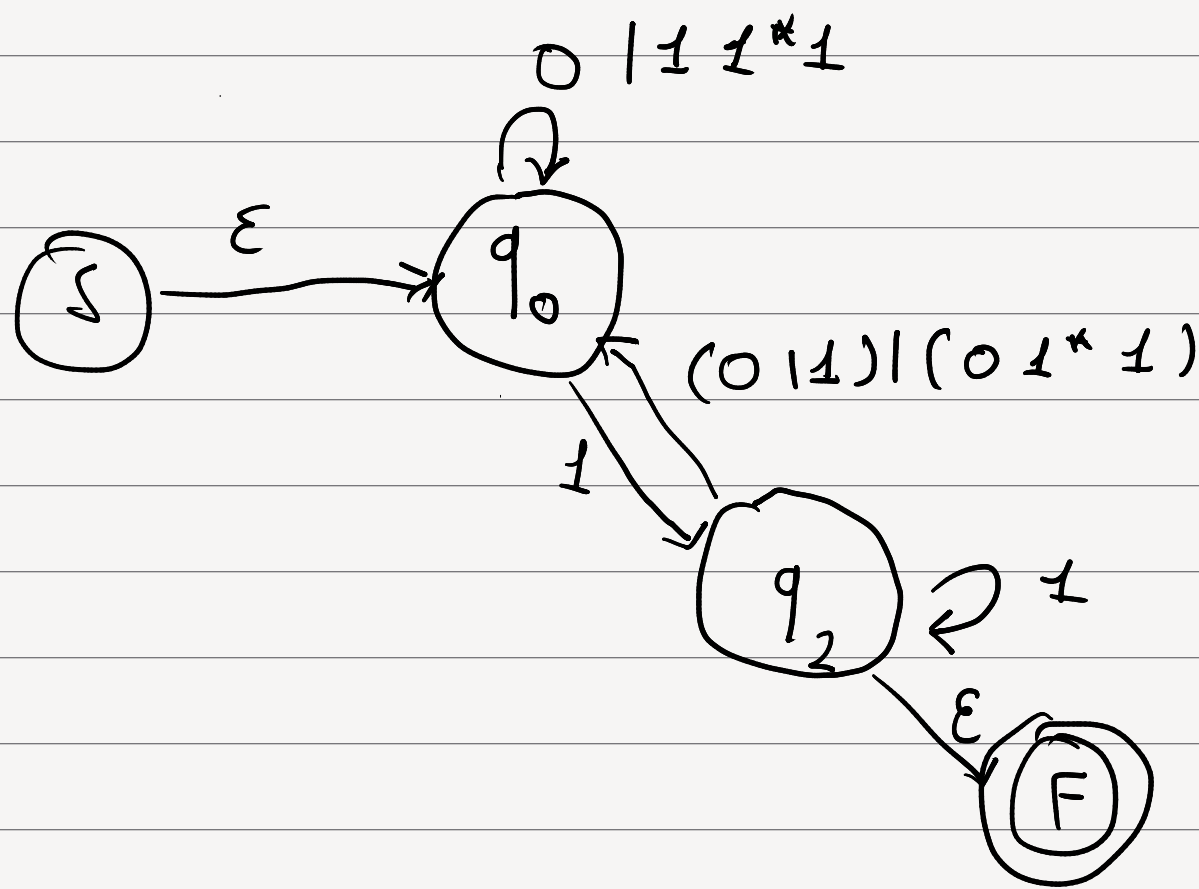
Determinare una grammatica di Tipo 3 che genera $T(M)$.



GNFA

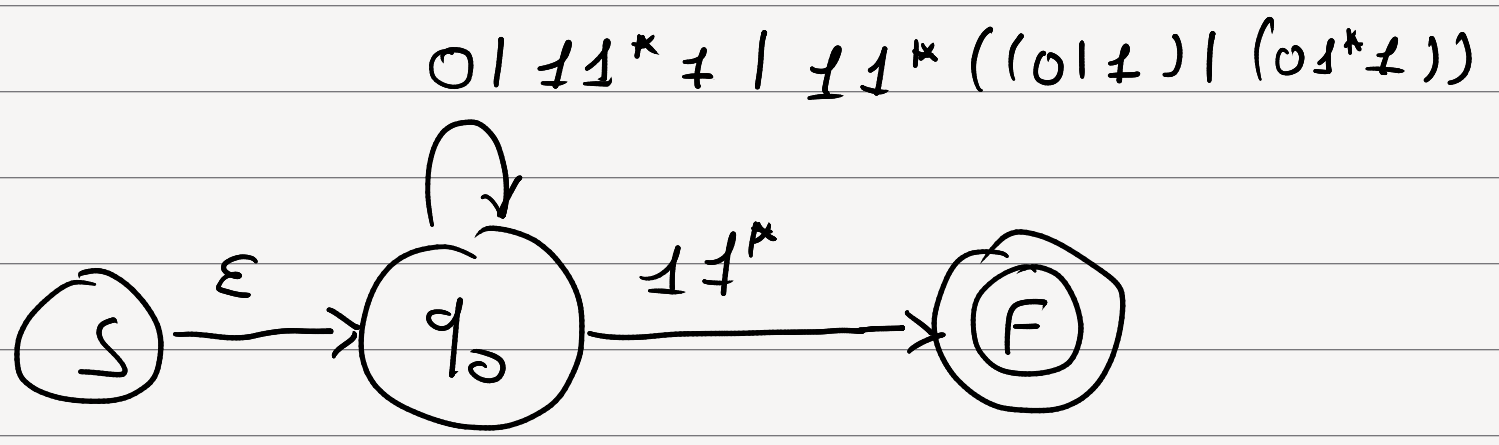


Togliamo q_1 :
 In Out
 q_0, q_2 q_0

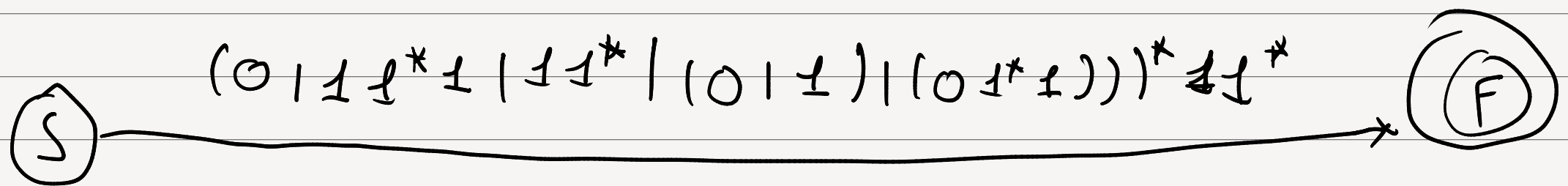


$q_0 \rightarrow q_0 = 1 1^* 1$
 $q_2 \rightarrow q_0 = 0 1^* 1$

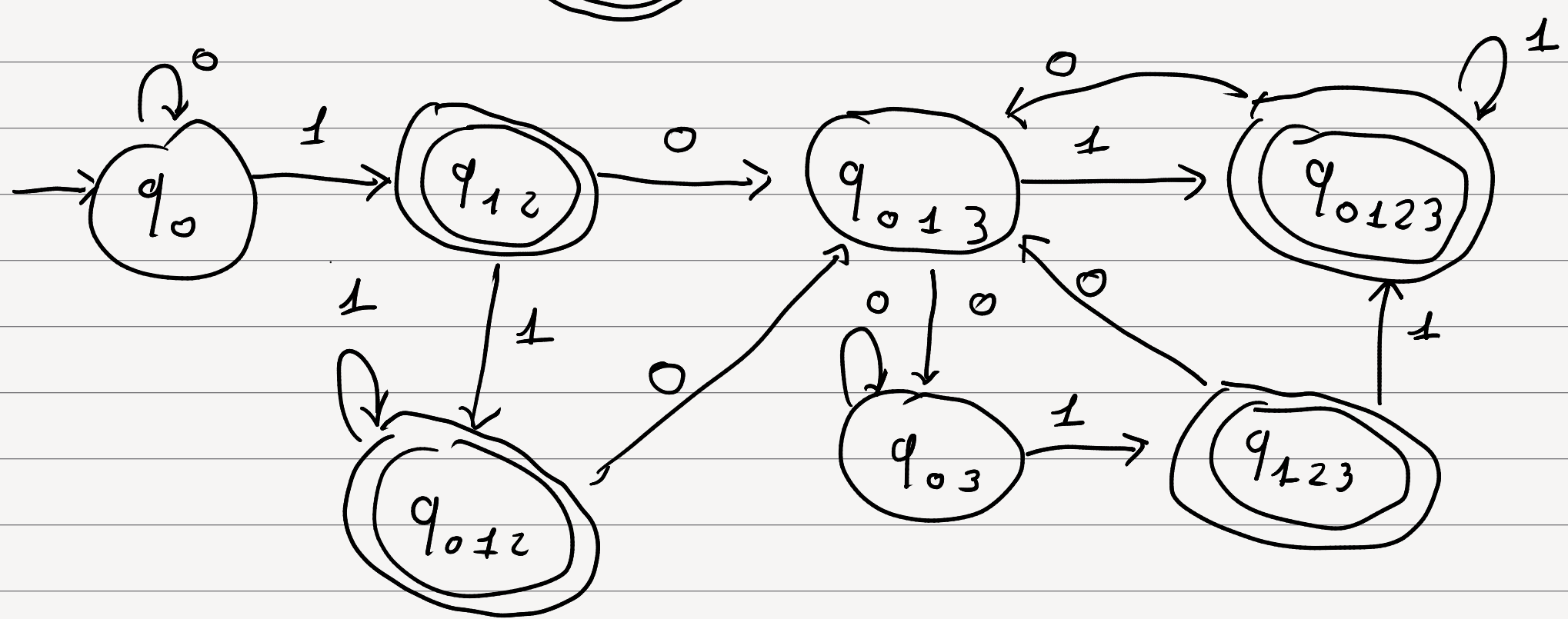
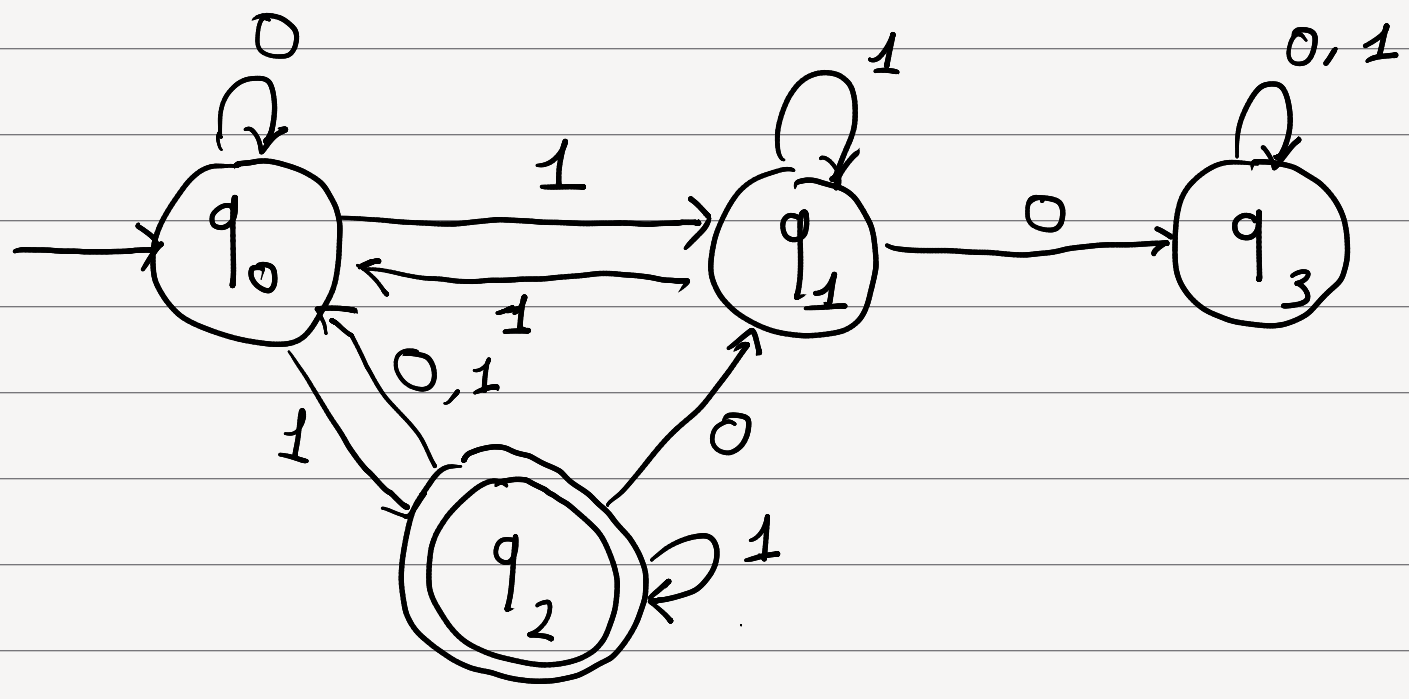
Togliamo q_2 :
 In Out
 q_0 q_0, F



$q_0 \rightarrow q_0 = 1 1^* ((0 1 1) | (0 1^* 1))$
 $q_0 \rightarrow F = 1 1^*$



Costruiamo il DFA:



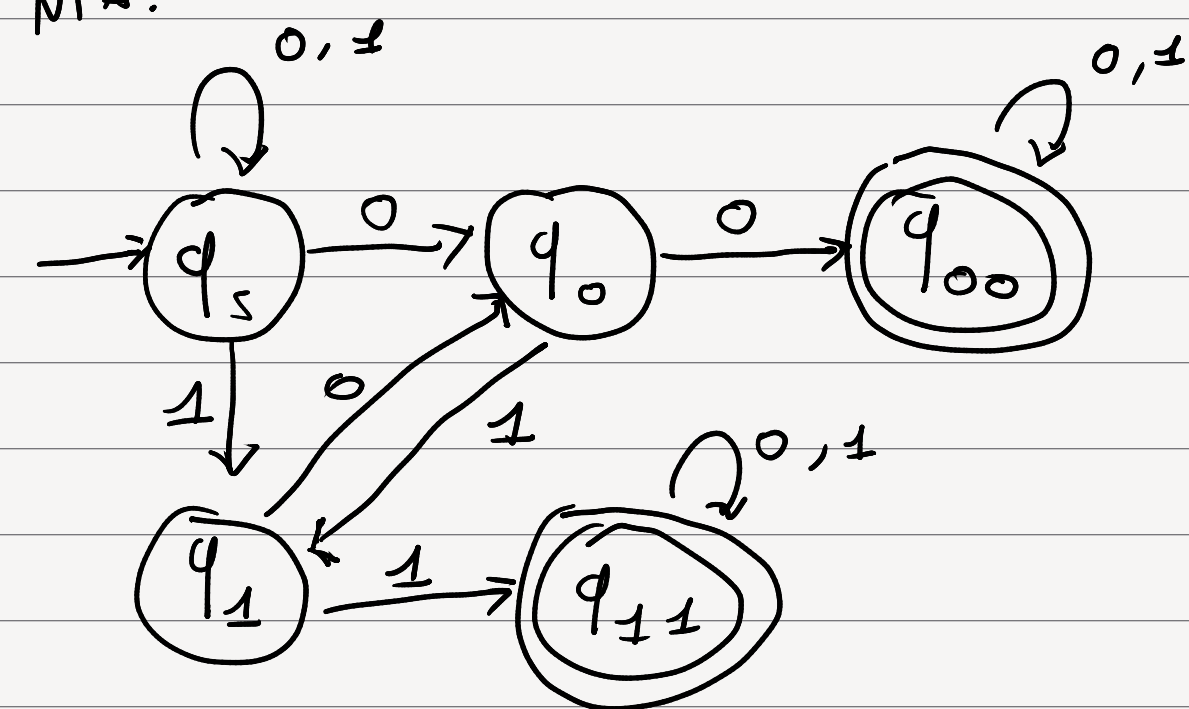
Esercizio 3

Sia dato il seguente linguaggio:

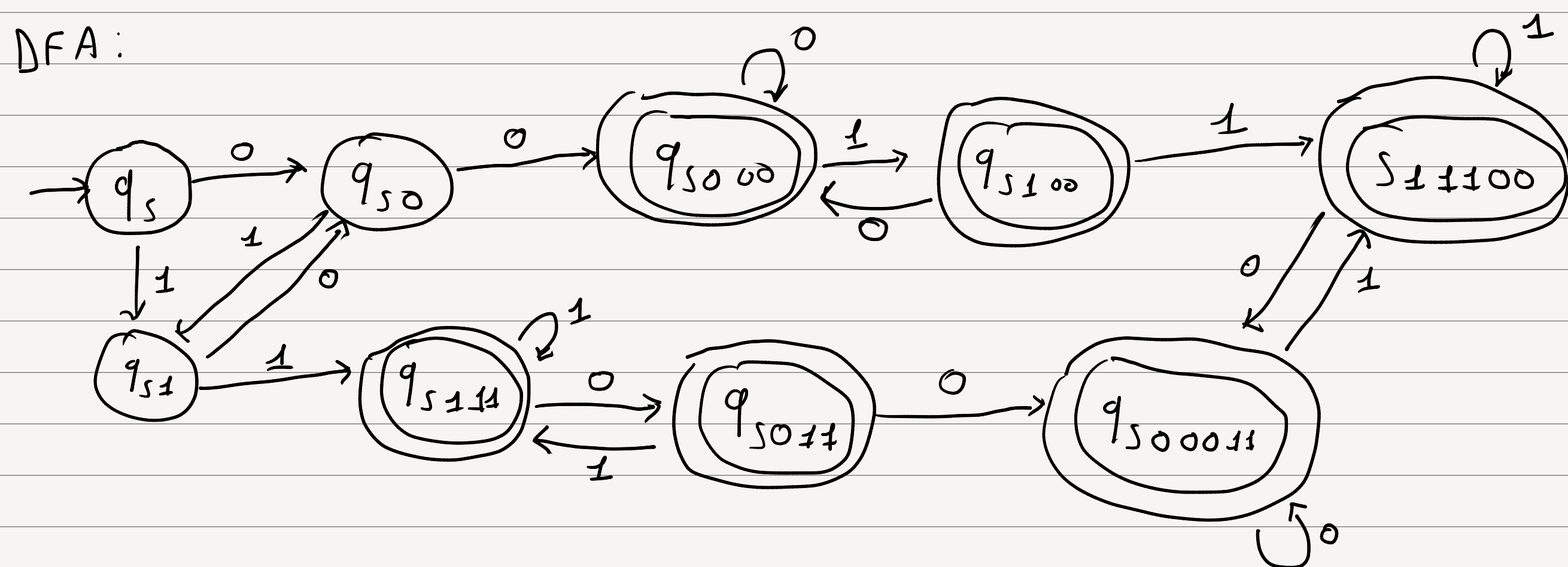
$$L = \{ \alpha 00\beta : \alpha, \beta \in \{0, 1\}^* \} \cup \{ \alpha 11\beta : \alpha, \beta \in \{0, 1\}^* \}$$

Costruire un automa a stati finiti deterministico che accetta L .

NFA:



DFA:

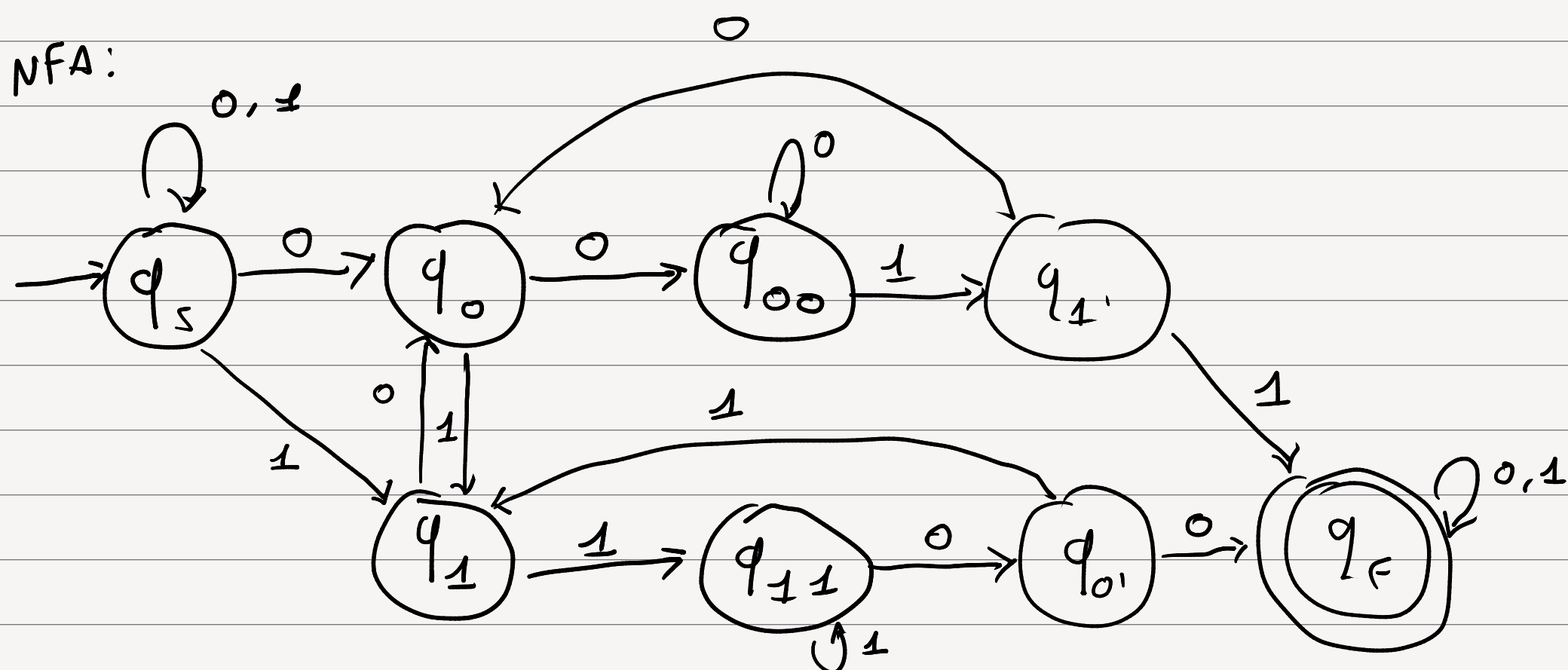


g) $\{ \alpha 00\beta : \alpha, \beta \in \{0, 1\}^* \} \cap \{ \alpha 11\beta : \alpha, \beta \in \{0, 1\}^* \}$

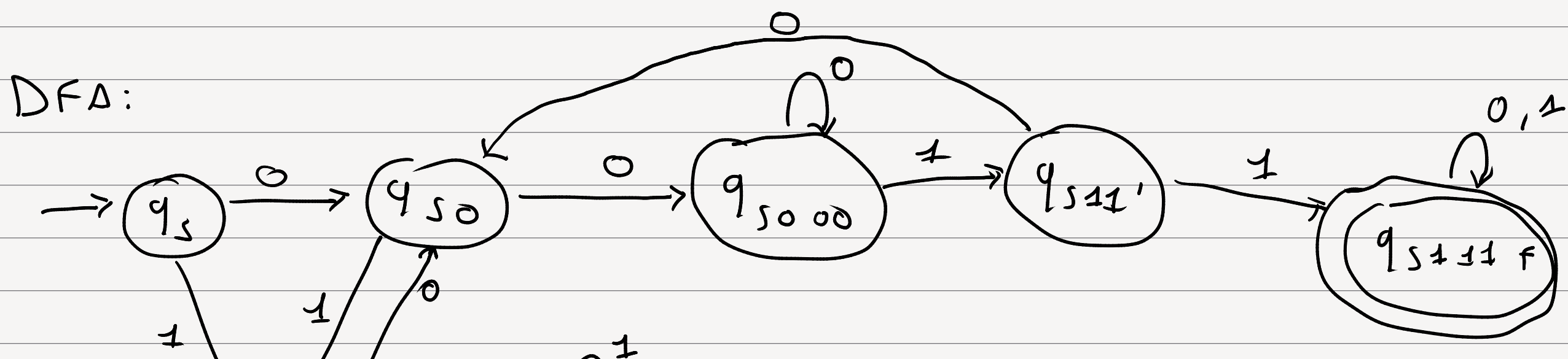
Costruire un automa a stati finiti deterministico che accetta L .

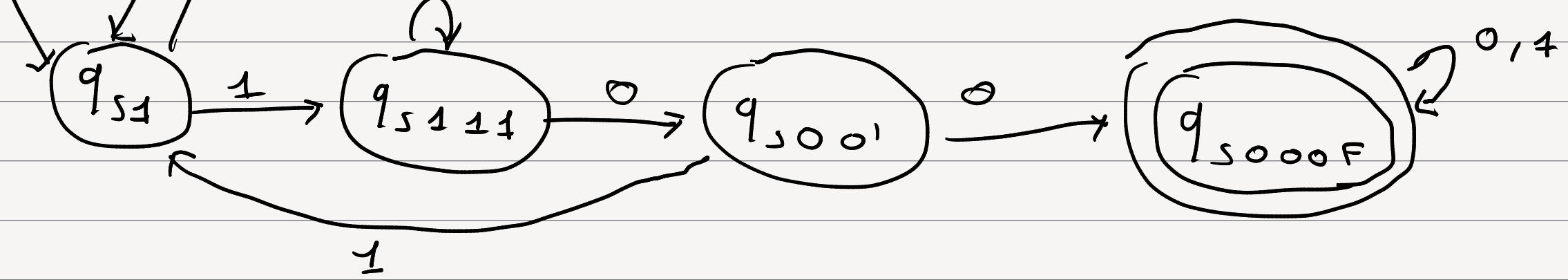
Determinare una grammatica lineare destra che genera L .

NFA:



DFA:





Esercizio 5

Sia dato il seguente linguaggio:

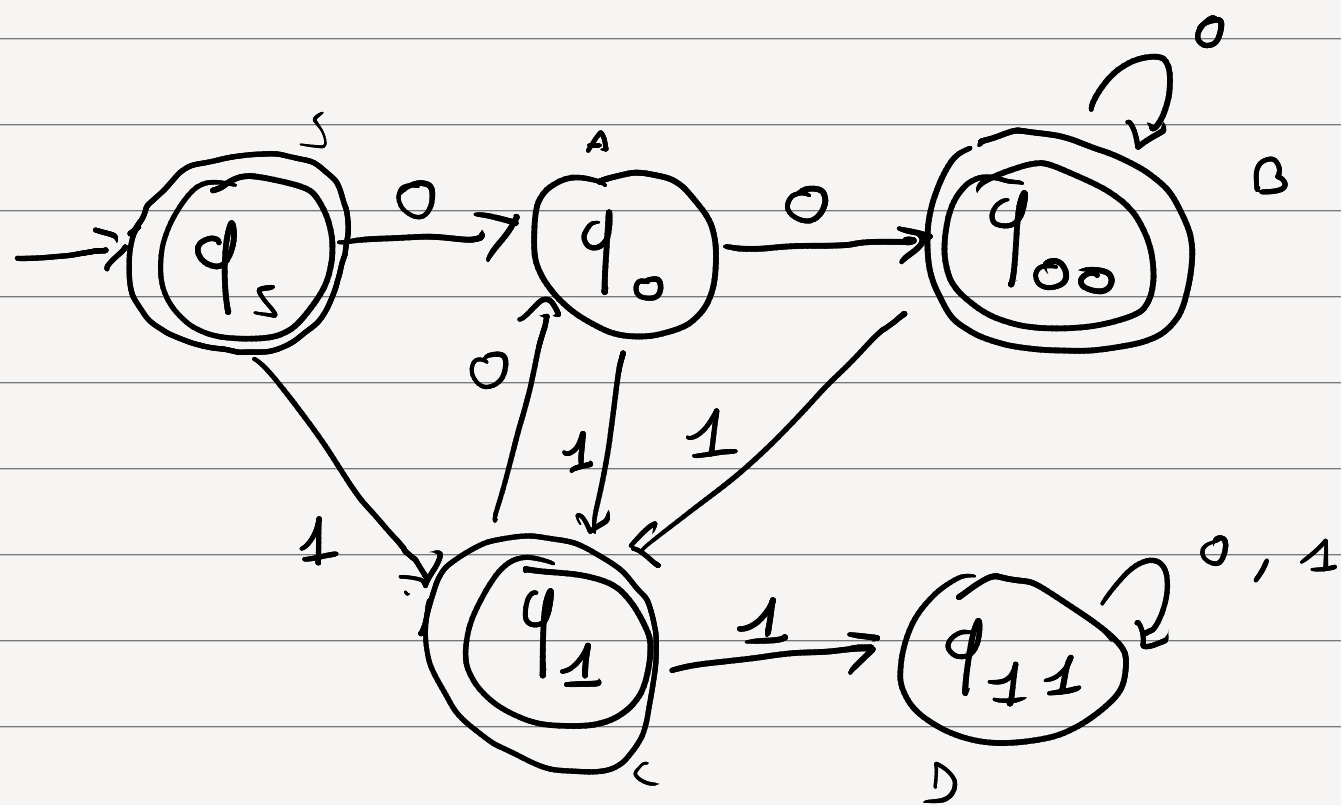
$$L = \{ \alpha 00\beta : \alpha, \beta \in \{0, 1\}^* \} \cup \{ w \in \{0, 1\}^* : w \text{ non contiene come sottostringa la parola } 11 \}$$

Costruire un automa a stati finiti che accetta L .

Costruire un automa a stati finiti deterministico che accetta L .

Determinare una grammatica lineare destra che genera L .

DFA:



CFG:

$$S \rightarrow 0A \mid 1C \mid 1 \mid \epsilon$$

$$A \rightarrow 0B \mid 1C \mid 1$$

$$C \rightarrow 0A$$

$$B \rightarrow 0B \mid 1C \mid 0 \mid 1$$

Esercizio 6

Sia L il linguaggio denotato dalla seguente espressione regolare su

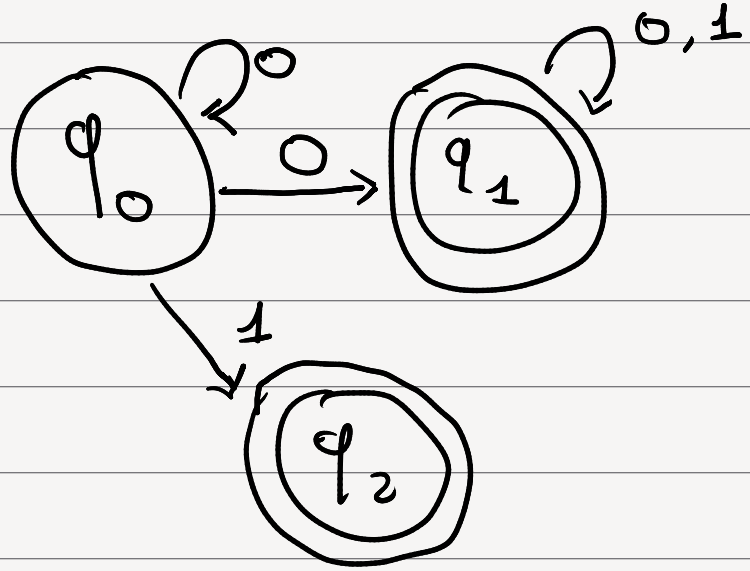
$$X = \{0, 1\}:$$

$$0(0+1)^* + 0^*1$$

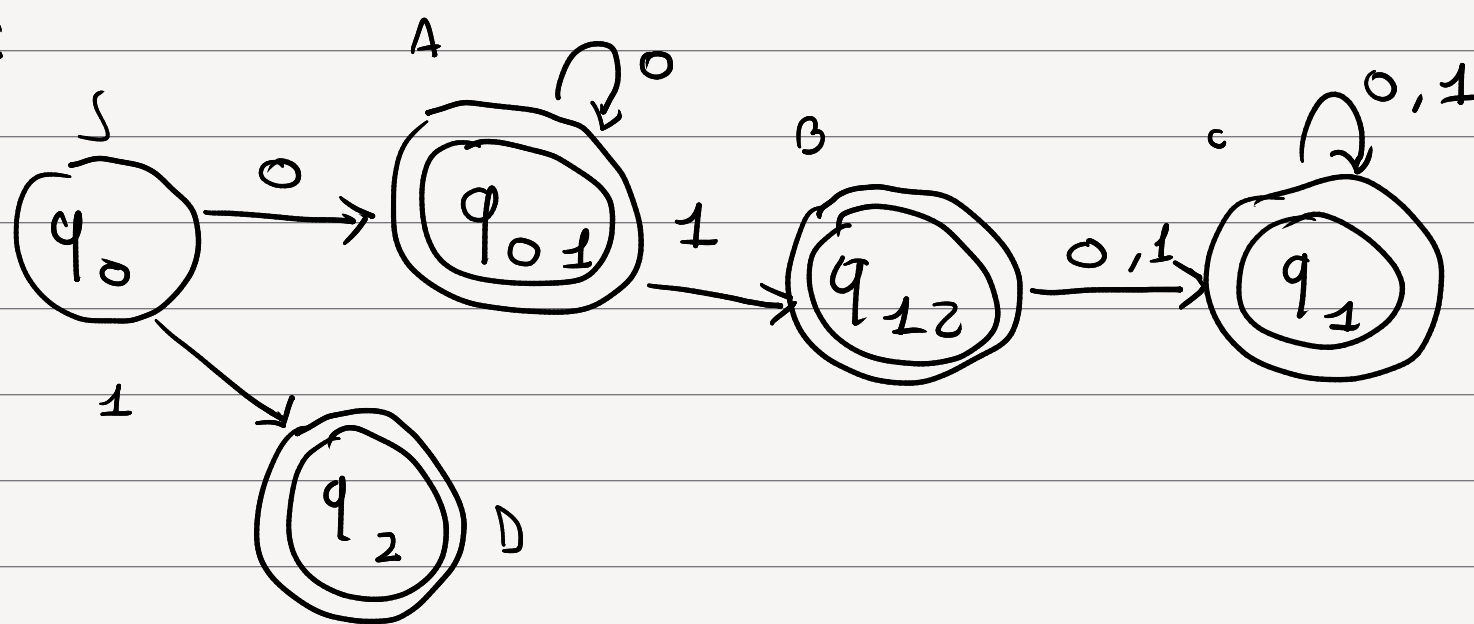
Costruire un automa a stati finiti deterministico che accetta L .

Determinare una grammatica lineare destra che genera L .

NFA



DFA:



CFG:

$$S \rightarrow 0A \mid 1D \mid 014$$

$$A \rightarrow 0A \mid 1B \mid 011$$

$$B \rightarrow 0C \mid 1C \mid 011$$

$$C \rightarrow 0C \mid 1C \mid 011$$

Esercizio 7

Sia data la seguente grammatica lineare destra:

$$G = (X, V, S, P)$$

ove $X = \{0, 1\}$

$$V = \{S, A, B\}$$

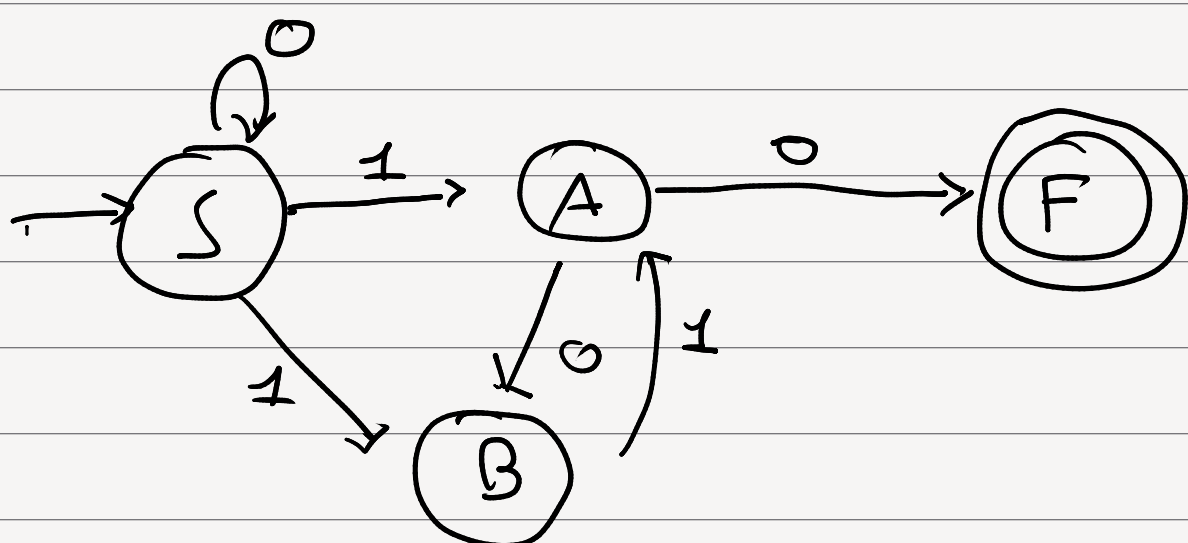
$$P = \{ S \rightarrow 0S \mid 1B \mid 1A,$$

$$A \rightarrow 0B \mid 0,$$

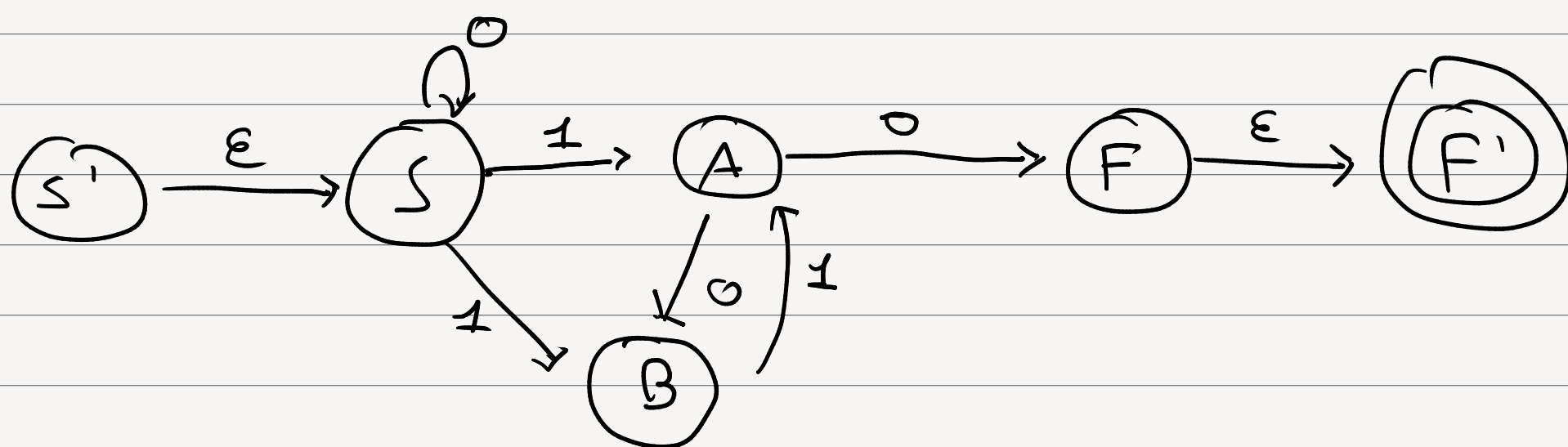
$$B \rightarrow 1A \quad \}$$

Determinare un'espressione regolare che denota $L(G)$.

Costruire un automa a stati finiti che riconosce $L(G)$.



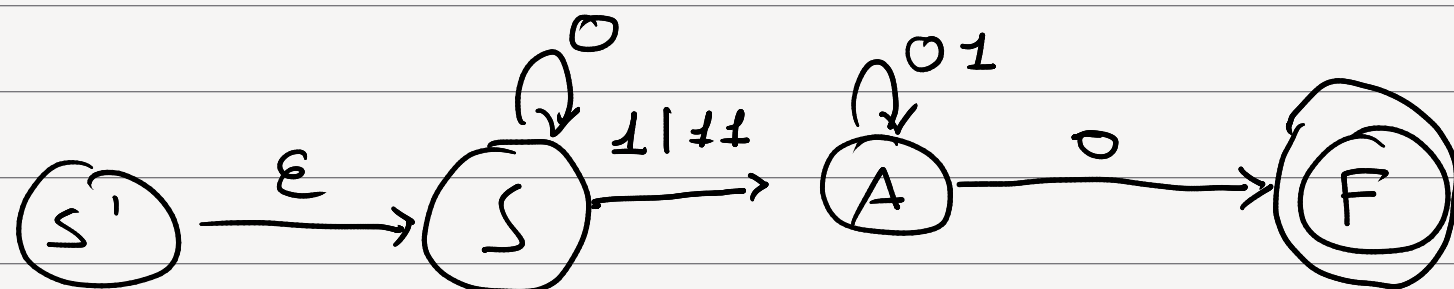
Costruiamo il GNFA:



Togliamo B

In Out

S, A A

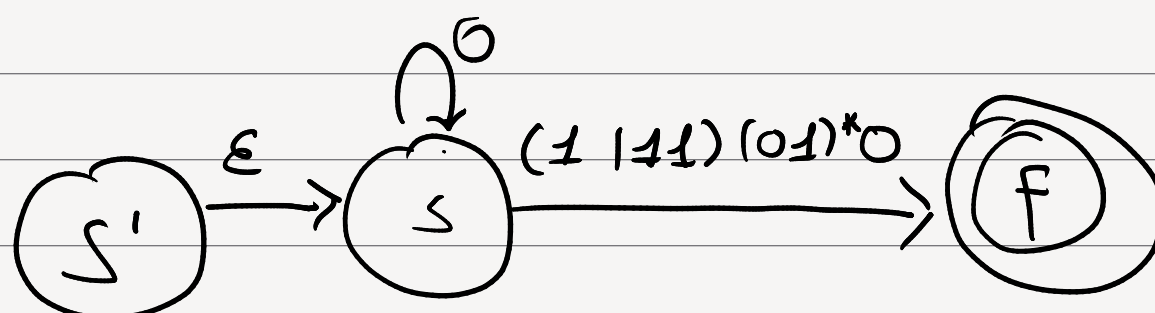


$S \rightarrow A = 11$

$A \rightarrow A = 01$

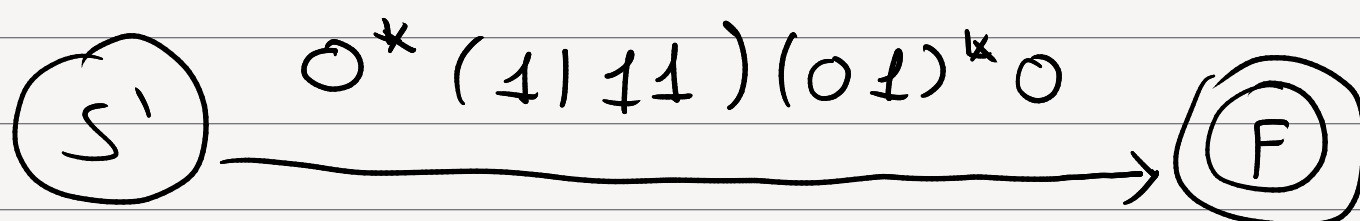
Togliamo A

In Out
S F



$S \rightarrow F = (11)(01)^*0$

Togliamo S:



Esercizio 10

Sia dato il linguaggio:

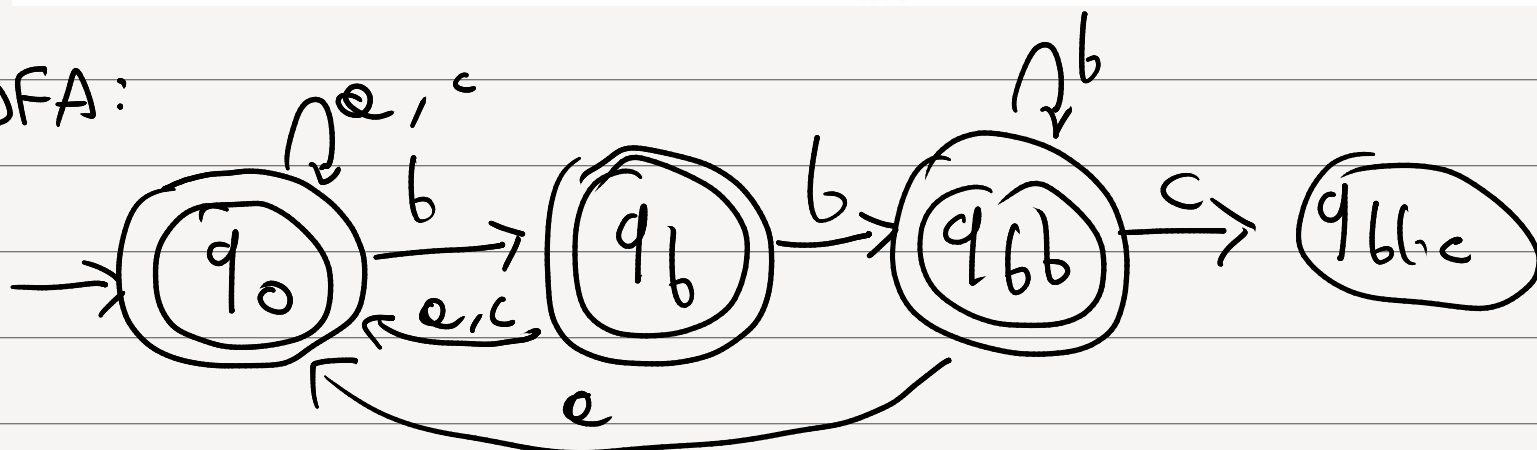
$L = \{ w \in \{a, b, c\}^* : w \text{ non contiene come sottostringa la parola } bbc \}$

Costruire un automa a stati finiti deterministico che accetta L .

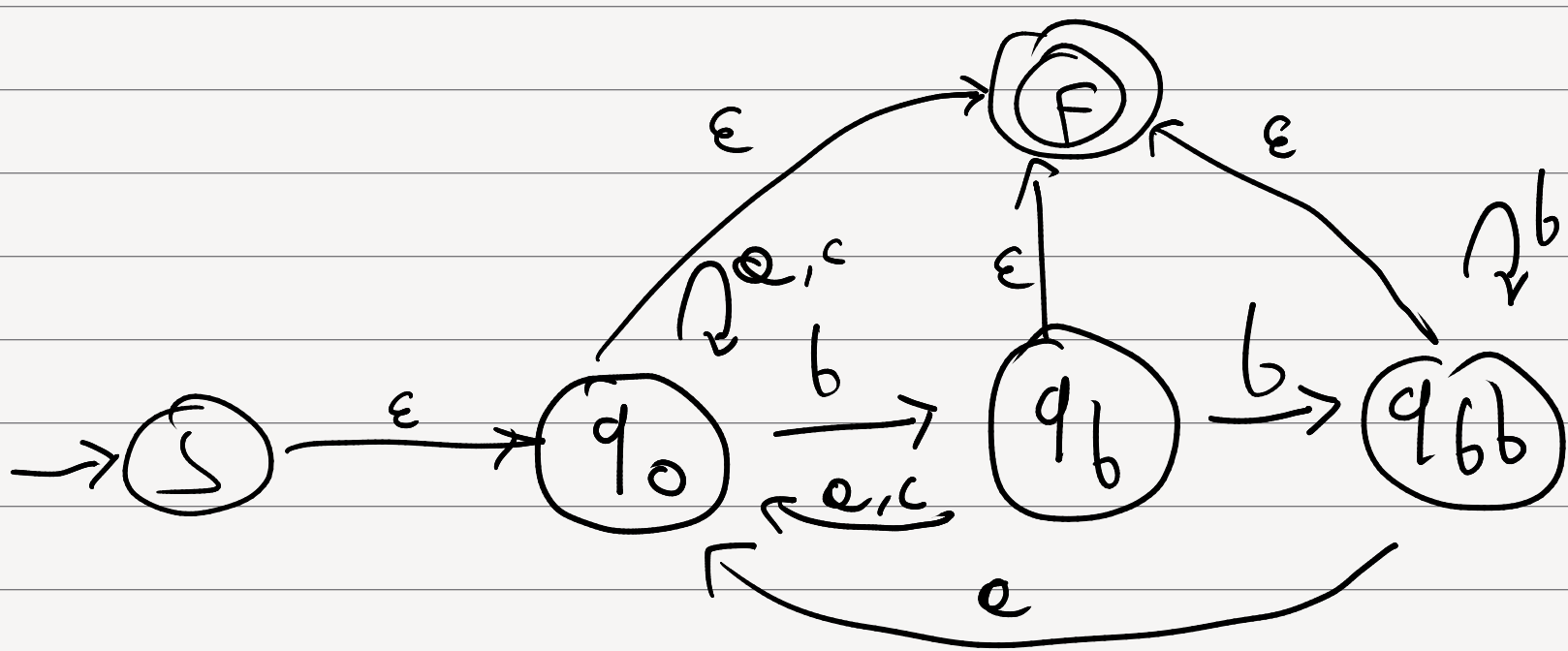
Determinare una grammatica lineare destra che genera L .

Determinare un'espressione regolare che denota L .

DFA:



Costruiamo il GNFA



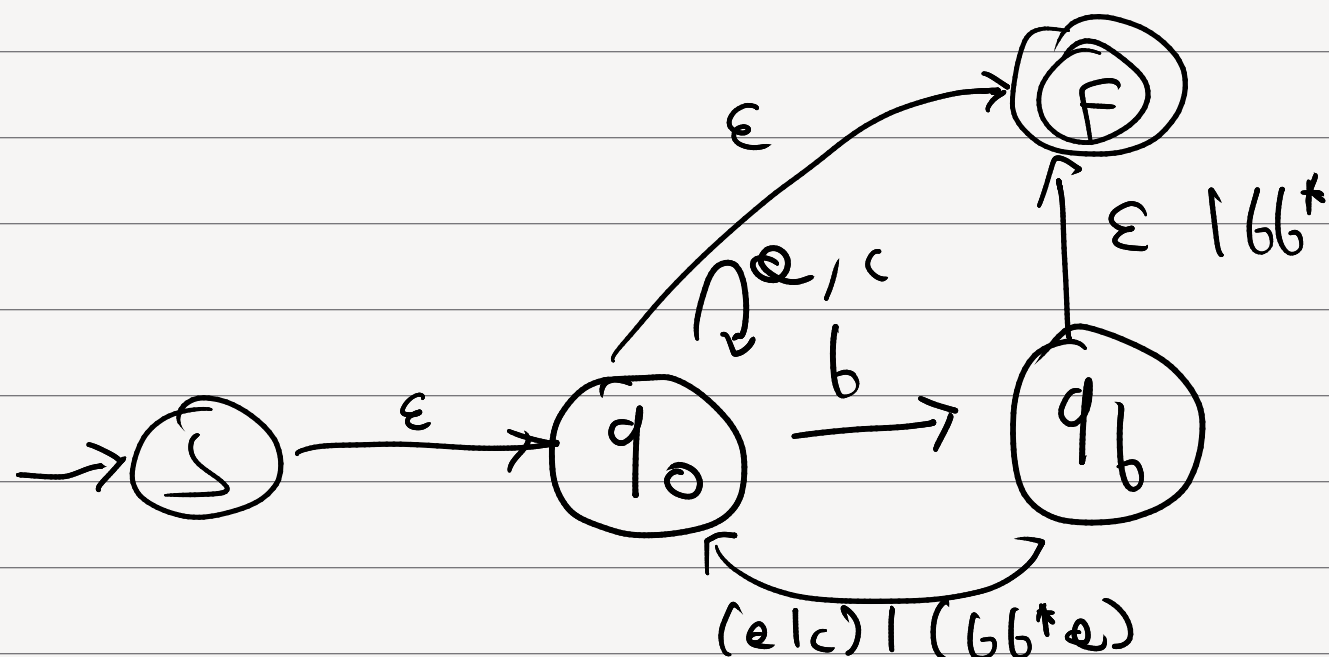
Elimino q_{1b}

In
 q_1

Out
 F, q_0

$$q_1 \rightarrow F = b b^*$$

$$q_1 \rightarrow q_0 = b b^* \epsilon$$



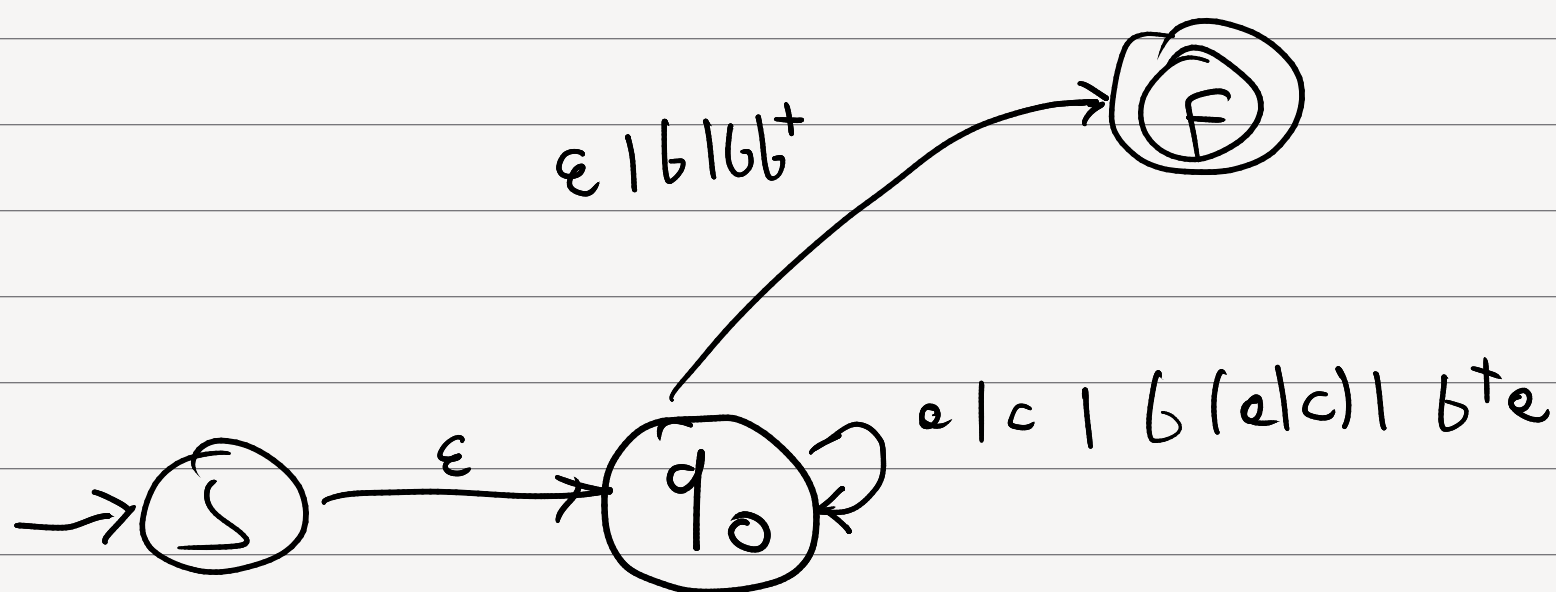
Elimino q_1

In
 q_0

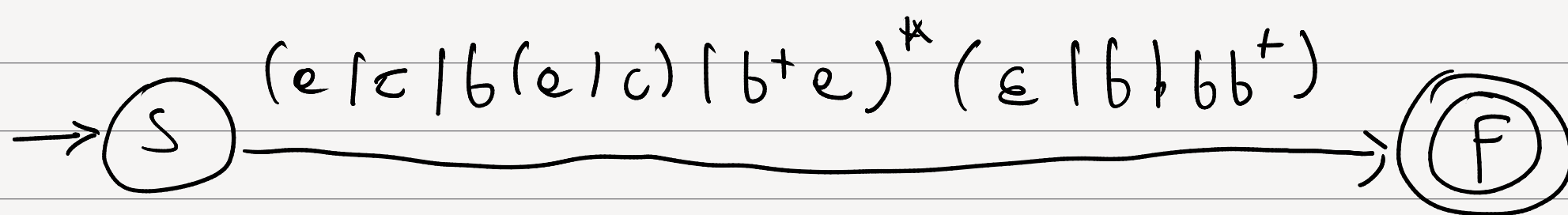
Out
 q_0, F

$$q_0 \rightarrow q_0 = b (\epsilon | c) | (b^+ \epsilon)$$

$$q_0 \rightarrow F = b | b b b^* = b | b b^+$$



Quindi:



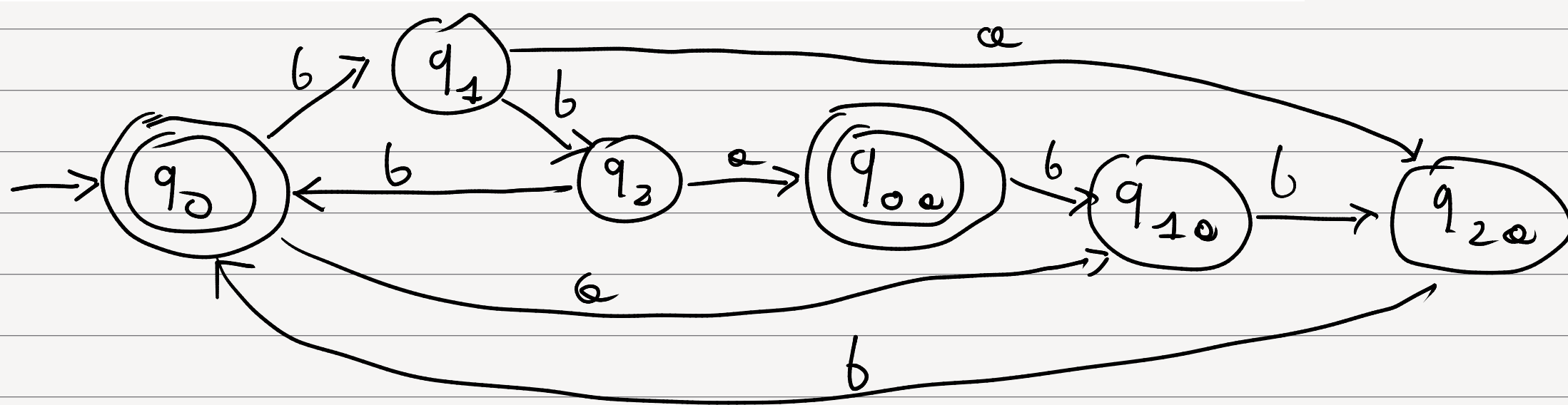
Esercizio 11

Sia dato il seguente linguaggio:

$L = \{ w \in \{a, b\}^* : |w| = 3k, k > 0 \text{ AND } w \text{ non contiene come sottostringa la parola } aa \}$

Costruire un automa a stati finiti che accetta L .

Determinare una grammatica lineare destra che genera L .



Esercizio 12

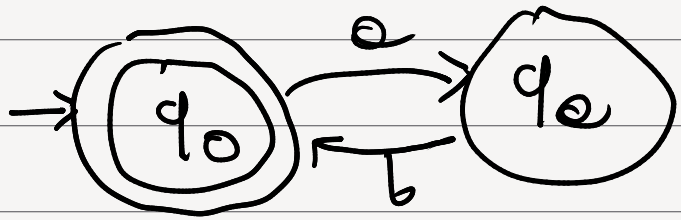
Siano L_1 ed L_2 i linguaggi denotati dalle espressioni regolari

$(ab)^*$ e $a(ba + a)^*$, rispettivamente.

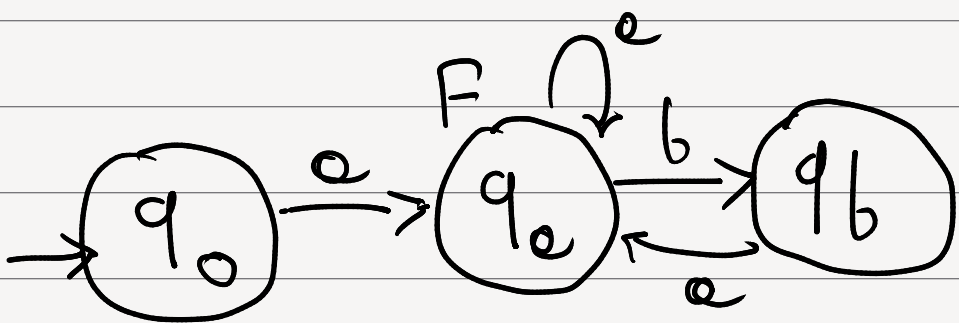
Costruire un automa a stati finiti che accetta il linguaggio

$$L = L_1 \cup L_2.$$

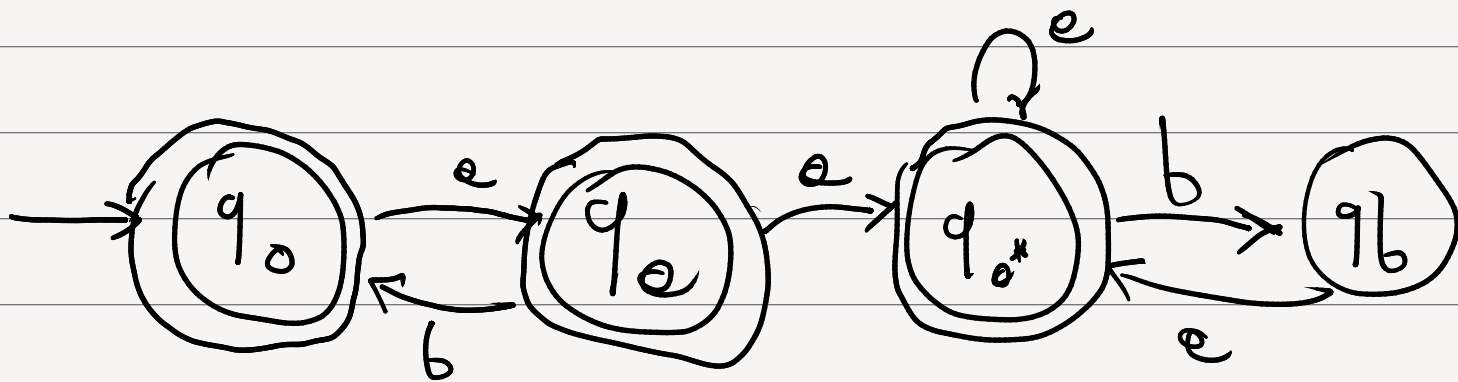
L_1 :



L_2 :



L :



5. Linguaggio delle stringhe che rappresentano numeri binari ($X = \{0, 1\}$) multipli di 3 (es. $0=0$, $11=3$, $110=6$, $1001=9$, ...)

